

ASSESSING SYMPTOM CHANGE OF NEURODEVELOPMENTAL AND
PSYCHOPATHOLOGY FACTORS USING LATENT GROWTH CURVE ANALYSIS IN
THE ABCD STUDY

BY

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Abstract

Diagnostic models such as the hierarchical taxonomy of psychopathology (HiTOP) are gaining notice amongst the psychopathology research community due to issues with traditional classification methods (Kotov et al., 2021). Yet, the placement of neurodevelopmental symptoms in HiTOP remains unclear, partly due to a lack of longitudinal research on their stability, and of their distinction from externalising symptoms. Using data from the Adolescent Brain Cognitive Development (ABCD) study, we conducted latent growth curve analysis to assess the comparable change of neurodevelopmental symptoms and other latent factors across pre-adolescence, using factor structure determined by Michelini and colleagues (2019). We predicted that neurodevelopmental symptoms would show the lowest improvement of model fit when constrained to represent change across time. We expected that this would also be reflected in greater associations with birth weight, due to functional developmental differences, in line with previous findings in this sample (Dooley et al., 2022). Findings indicated that while the specification of a slope (constrained change of symptom severity) for the neurodevelopmental factor fit the data less well than a slope for the internalising and externalising factors, these differences were only moderate. After controlling for covariates, we found no relationship to be significant between birth weight and any psychopathology domain, with pre-term birth appearing to be a strong and unique predictor of neurodevelopmental symptoms, and household income being the strongest predictor of internalising and externalising symptoms. These findings implicate both the distinctiveness of neurodevelopmental symptoms and the effect of the p factor, which represents co-occurrence of symptoms in general. Future research should continue to investigate the placement of neurodevelopmental symptoms within psychopathology factor models.

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Introduction

The Status of Psychiatric Nosology

In recent years, considerable research on Psychiatric Nosology has questioned the validity and homogeneity of categorical classifications in the Diagnostic and Statistical Manual for Mental Disorders (DSM) and International Classification of Diseases (ICD), which are the predominant systems used for diagnosing psychopathology around the world (Reed et al., 2011). Categorical systems like the DSM and ICD were first developed to provide clinicians with standardised measurable criteria which could be used to determine treatment course (Blashfield et al., 2014; Balling et al., 2023), in a similar manner to the way in which physical illnesses are diagnosed (Clark et al., 2017). As suggested by Reed and colleagues (2013), these systems can be seen as a useful way to characterise and treat different types of mental health symptoms, as they delineate the presence of abnormal behaviours into specific, measurable criteria and provide diagnostic thresholds so only severe cases (i.e., where impairments limit normal everyday functioning) are provided with a diagnosis. Some of the disorders in the DSM (e.g., major depressive disorder (MDD), panic disorder, bipolar disorder, and schizophrenia) are suggested to have decent construct validity (Lilienfeld & Treadway, 2016). However, unlike most physical illnesses, diagnostic criteria for mental illnesses have largely been determined by behavioural observations and theoretical concepts, rather than biological or aetiological evidence (Clark et al., 2017), which has resulted in heavy criticism regarding the application of such classifications for clinical use (Hoff et al., 2020; Kotov et al., 2017). For instance, Kendler (2017) argues that categorical systems represent a reification of abnormal behaviour which does not yet have well-founded aetiological pathophysiology.

While categorical differentiation is necessary to determine which patients require treatment, such diagnoses have resulted in public perception that mental disorders are

inherently categorical in nature, enabling misconstrued ideas about what the diagnosis of a disorder represents (Meyer et al., 2020), which can result in public stigma towards mental health. Furthermore, a study by Thibodeau (2020) indicates that most stigma towards psychopathologies are largely related to the commonly held view that disorders are discrete conditions. It is worth mentioning that despite the effect that categorical interpretations have on stigma, there are indications that some western populations are becoming more aware of the continuous nature of psychopathologies, although this is more noticeable for common diagnoses, such as MDD, than for less common diagnoses, such as schizophrenia (Angermeyer et al., 2015). Yet, while the DSM has been updated with several revised editions since its conception, the use of categorical criteria remains embedded in the diagnostic process. For example, in the fifth edition of the DSM (DSM-5), to receive a diagnosis of Attention Deficit-Hyperactivity Disorder (ADHD), a child under the age of 17 must display at least 6 or more symptoms of hyperactivity, impulsivity, or inattention (Childers & Bass, 2019); The number of symptoms, among other criteria, being the distinctive measurable features that are used to differentiate a child that ‘has ADHD’, from a child that does not (see Antshel et al. (2013) for further information regarding diagnostic criteria of ADHD). Thus, current diagnoses are based on polythetic criteria, rather than an inherent presentation of the condition, which has been suggested by Balling and colleagues (2023) to result in the loss of unique information about a patient’s symptom presentation. Furthermore, a study by Bickman and colleagues (2012) which investigated the reliability and validity of clinical diagnoses used in mental health services demonstrated several issues with the current classification system, such as poor internal consistency, low inter-rater reliability, and differences in diagnosis rates between sites. The latter issue has been found cross-culturally too (Mesman et al., 2016).

It has been suggested that symptoms attributed to most mental disorders can be more accurately conceptualised as continuous spectra which encompass varying levels of severity, rather than discrete behavioural abnormalities that are caused by an underlying condition or impairment (Van Os et al., 2023; Shankman et al., 2018; Borsboom et al., 2016; Kotov et al., 2017; Fair et al., 2012; Bala et al., 2021; Snyder et al., 2022; Kotov et al., 2021; Newson et al., 2021; Chamberlain et al., 2018). Such evidence comes from an era of psychological science which is becoming increasingly data-driven and open-minded about the constructs that it aims to measure (Hoff et al., 2020). Subsequently, this line of questioning has helped establish that diagnostic criteria do not necessarily represent homogeneous groups or conditions, as those who meet criteria for the same diagnosis may present with largely different symptoms or severity, a phenomenon also known as heterogeneity. Heterogeneity is particularly noticeable in ADHD (Lilienfeld & Treadway, 2016), due to the 18 different symptoms that can be used to form a diagnosis using the DSM-5 (Antshel et al., 2013), despite only 6 being required to meet the diagnostic threshold (Childers & Bass, 2019). Additionally, such symptoms tend to fluctuate throughout the lifespan, making it difficult to assess a person accurately at a given timepoint (Caspi et al., 2014). Thus, due to the continuous nature of most symptoms, and the heterogeneity that comes with most diagnoses, questions have arisen about whether diagnosing patients categorically is the best way to provide efficacious treatment outcomes for patients. Two options have subsequently been frequently debated over the years: (1) Continue to use and improve categorical systems for their standardised utility, or (2) Shift to a dimensional approach in the search for conceptual validity.

Psychopathology as a Hierarchical Nosology

Much research on psychopathology in recent years has aimed to determine meaningful factor structures that could inform frameworks of nosology. In this area, multiple

studies indicate that psychopathology is best described as a hierarchy of symptom specificity and severity, whereby highly correlated symptoms form specific dimensions, but symptoms are still correlated across dimensions. For example, it is now well established in the literature that psychopathology is best represented at its broadest by a general factor, which accounts for correlations across most (if not all) measures of mental health (Lahey et al., 2012). Coined the '*p factor*' by Caspi and colleagues (2014), the general factor of psychopathology has helped advance understanding of the relationship between symptom specificity and symptom severity. According to Caspi and colleagues (2014), the p factor encompasses most (if not all) measures of mental health and well-being. It is associated with worse developmental history (e.g., childhood adversity, trauma), greater familial and genetic risk for symptoms, and more life in impairment. Other findings indicate that the p factor is also associated with cognitive issues, such as deficits in executive functioning (Romer & Pizzagalli, 2021; Hatoum et al., 2018a; Bloemen et al., 2018) and other general measures of cognition (Abramovitch et al., 2021), which are noticeable as early as age 3 (Caspi et al., 2020), suggesting that the p factor may be transdiagnostic to current clinical concepts, reflecting a more generalised representation of mental and cognitive function. To add to the empirical validity of the p factor, its presence has been well replicated in other studies using factor analysis (Martel et al., 2017; McElroy et al., 2018; Laceulle et al., 2015; Brislin et al., 2022; Romer et al., 2018; Castellanos-Ryan et al., 2016; Lahey et al., 2018; Patalay et al., 2015). These findings are sustained when assessing its presence longitudinally as well (Caspi et al., 2014; Romer & Pizzagalli, 2021; Caspi et al., 2020). The p factor is thought to reflect why there has been little progress in elucidating disorder-specific aetiologies, biomarkers, and treatments for categorically classified disorders (Caspi et al., 2014; Kotov et al., 2021)

Expanding on the conceptualisation of the p factor, studies involving factor analysis have established hierarchical structures which span from specific problems to broader

correlations of specific groups of symptoms, and then to the overarching p factor, which accounts for correlations across all symptoms. For instance, Michelini and colleagues (2019) conducted exploratory factor analysis (EFA) to assess the structure of the child behaviour checklist (CBCL) and adult self-report (ASR) from the Achenbach System of Empirically Based Assessment (ASEBA) using baseline data of 9 to 10-year-old children and their parents/caregivers from the ABCD study. The findings from this study indicated that both the CBCL and ASR are best characterised by the presence of the p factor, and five sub-factors, of which each represent broader symptom constructs. In the childhood analysis, factors were semantically labelled as Internalising, Externalising, Somatoform, Detachment, and Neurodevelopmental. These sub-factors can be thought of as representing the middle tier of a psychopathology hierarchy, whereby each sub-factor represents a set of correlated symptoms, of which are also correlated (to a lesser extent) with symptoms in other sub-factors, which represents the p factor. Furthermore, the p factor can be thought of as the cumulative interaction between symptom severity and comorbidity, evidence for which is further extended across multiple studies (Papachristou & Flouri, 2020; Winsper et al., 2020; Van Os et al., 2014; Haltigan et al., 2018) that identify the presence of a psychopathology interactome, whereby the presence of severe symptoms are likely to play a part in causing the presence of other symptoms as well (Van Os et al., 2014), a relationship which is established as being persistently maintained across the lifespan (Caspi et al., 2020). Latent class analyses have also helped to demonstrate the relationship between symptom severity and symptom co-occurrence, whereby people assigned to the most severe class are likely to have symptoms which are more severe across all dimensions of psychopathology (Basten et al., 2013; Bianchi et al., 2017; Retzler et al., 2023). While there do appear to be groups of people who cluster together based on specific symptoms, the gradient of symptoms from non-existent to extremely severe can be thought of as a spectrum, whereby increased severity is likely to

result in comorbid symptoms, and lesser severity is likely to result in mild, isolated symptoms. As further demonstration of this relationship, clustering analyses demonstrate that when grouped together based on symptoms, groups do not tend to form primarily based on diagnosis (Heckers et al., 2013; Kushki et al., 2019; van der Meer et al., 2012; Chavez-Baldini et al., 2023). A noticeable trend in these analyses is that groups tend to cluster based on symptom severity, suggesting that individuals with severe symptom presentations are more likely to have comorbid symptoms (and a higher p factor).

The structure of sub-domains related to the p factor are not purely based on correlations though, for instance, the behavioural dysregulation profile (BDP) is thought to represent functional differences in the regulation of emotion and behaviour, aligning with the HiTOP model of correlated symptoms, due to the presence of both specific and underlying factors. Subsequently, the BDP is thought to act as a clinical proxy for the p factor (Aitken et al., 2019; Sloan et al., 2017), which has been supported by factor analysis (Haltigan et al., 2018; Miller et al., 2019). Furthermore, some groups of symptoms can be found to share associations with functional impairments, whereas others may be more broadly related because the presence of such symptoms may influence the development of others. Yet, while each diagnosis in the DSM aims to represent a discrete behaviour or function, most studies assessing the relationship between a hypothesised risk factor (or aetiology) with a specific diagnosis have helped establish that in most cases, regardless of diagnosis, there are typically a multitude of biological, genetic, environmental, and social factors which are present, all interacting to influence the development of symptoms (Clark et al., 2017; Hankin et al., 2015; Lahey et al., 2017). This evidence provides support against the historically held view that each diagnosis must have a specific and/or unique cause (Clark et al., 2017). Some of the most well-covered known risk factors for general psychopathology include childhood adversity (Mamun et al., 2023), poverty (Lund et al., 2010), domestic violence (Harold &

Sellers, 2018), genetics (Martin et al., 2018; Smoller et al., 2019), paternal and maternal mental health problems (Middeldorp et al., 2016), and pre-natal exposures (e.g., alcohol, tobacco, marijuana, cocaine) (Roffman et al., 2021), although there are many others that could be included here as well. Such findings tend to suggest that these factors are cumulatively influential in determining psychopathology (Lahey et al., 2017), and that they are typically all associated with symptoms in general (i.e., the p factor), rather than any specific diagnosis (Caspi et al., 2014). While common risk factors do appear to affect most psychopathologies in general, there is evidence for certain types of symptoms to have more homogeneous aetiologies, such as the high genetic component associated with autism spectrum disorder (ASD) (Yip et al., 2018).

A hierarchical structure of psychopathology (i.e., correlations between symptoms and dimensions) is further supported by evidence for shared aetiologies between different types of symptoms. For example, much of the research on psychopathology since the conception of diagnoses has been to elucidate biomarkers, or specific aetiological functions that can be targeted for treatment and prevention purposes. However, findings from the large volume of studies assessing neurobiology (Vandewouw et al., 2023; Mewton et al., 2022; Jacobs et al., 2021; Kushki et al., 2021; Kushki et al., 2019; Schettini et al., 2021; Romer et al., 2018) and genetics (Ronald et al., 2008; Cordova et al., 2022; Clarke & Eapen, 2021; Brikell et al., 2020; Dajani et al., 2016; Huang et al., 2021; Ghirardi et al., 2021; Smoller et al., 2019; Martin et al., 2018; Loughnan et al., 2022) indicate that there is little evidence in the neuroscience field that delineates mental disorders from each other, with the most noticeable neurobiological abnormalities being associated with general mental health issues, rather than any specific symptom or disorder. Much of the literature in this area suggests that this observation is further explained by genetic effects on mental health symptoms which are pleiotropic (Martin et al., 2018; Martin et al., 2015; Vorstman et al., 2017), and that this may

also be the case for neo-natal environmental risk (Dooley et al., 2023a; Zhang et al., 2022) and social environmental adversities too (Clark et al., 2017; Lahey et al., 2017). For example, additive forms of childhood maltreatment have been found to be associated with increased comorbidity of psychiatric disorders and worse well-being outcomes in general, which is in turn mediated by genetic factors (Lippard & Nemeroff, 2020; Smoller et al., 2019). Additionally, findings from Vrijnsen et al. (2018) suggest that childhood trauma can have a cumulative effect on the likelihood of comorbid symptoms later in life. It is worth noting that this does not mean each risk factor or gene implicated has the same influence on a given symptom, or on broader dimensions either. A study by Waszczuk and colleagues (2021) found some polygenic scores to be more specifically related to neurodevelopmental and externalising symptoms than others. Thus, these findings represent the capacity for different symptoms to be both shared and discrete regarding aetiology.

Due to categorical systems of diagnosis (e.g., DSM, ICD) being the predominant method globally, and with a lack of comparative systems available, they have resulted in heavy criticism. Possibly the most widely used critique of categorical frameworks for psychopathology is the evidence for high rates of co-occurring diagnoses, also referred to as comorbidity (Mersin Kilic et al., 2020; Gair et al., 2021; Brown & Barlow, 2009; Leichenring et al., 2023; Clark et al., 2017; Hatch et al., 2023; Antshel et al., 2013). Comorbidity is partly explained by the research on the hierarchical structure of symptoms and the p factor, as mentioned above. The issue of comorbidity directly challenges the assumption made by diagnosis within a categorical framework that the criteria for a disorder represents a discrete condition. While in the cases referenced above this has been suggested to be a problem, others argue otherwise. For instance, Wittchen & Beesdo-Baum (2018) suggest that comorbidity is less a problem with the diagnostic method, but a way of observing the presence of different conditions, since giving a diagnosis of one disorder does not require

the clinician to refrain from assigning a separate, additional diagnosis as well. However, there is evidence to suggest that comorbidities appear because of shared causal attributes, rather than random co-occurrence. Those who receive a diagnosis typically have more severe symptoms than non-diagnosed patients, not just within their diagnosed condition, but among other conditions for which they do not reach the diagnostic threshold as well (Antshel et al., 2013; Caspi et al., 2020). Secondly, there is evidence to suggest that comorbidity is a representation of different symptoms being attributed to similar neurophysiological impairments (Pasion et al., 2023; Mewton et al., 2022; Jacobs et al., 2021; Kushki et al., 2019), which may explain why many diagnoses share associations with biological and genetic abnormalities. Additionally, aligning with these findings, the p factor has also been associated with smaller global brain volume and surface area, highlighting neurobiological markers of general mental impairment (Romer et al., 2023). Also, comorbidities are associated with worse transdiagnostic outcomes, such as quality of life and peer problems (Armstrong et al., 2015), which are likely involved interactively with mental health as well.

HiTOP

In search of a framework which incorporates the findings mentioned in previous sections, a consortium of researchers has been collaborating to develop an alternative, dimensional model for assessing psychopathology, called The Hierarchical Taxonomy of Psychopathology (HiTOP) (Kotov et al., 2017). HiTOP is still under development and is open to structural changes should new evidence emerge (Kotov et al., 2021), but it is fundamentally different from traditional diagnostic methods, which has resulted in notable support (Haslam et al., 2020; Newson et al., 2021; Conway et al., 2019), alongside notable criticism (Wittchen & Beesdo-Baum, 2018; Tyrer, 2018; Zimmerman, 2021; Haefffel et al., 2022). Possibly the most relevant criticism is related to the perceived lack of improvement to patient outcomes, whereby the HiTOP framework may address the conceptual issues with

classification, but it does not solve the overarching issue of heterogeneity and multifinality of both aetiology and symptoms themselves (Komasi et al., 2023). There is initial evidence to suggest that the HiTOP model would provide better clinical utility than the DSM, from a study which asked clinicians to rate different components of each system (Balling et al., 2023), and from a meta-analysis indicating that the framework fits with the literature consensus on psychopathology factor structure (Ringwald et al., 2023). Additionally, the model aims to be inclusive cross-culturally, and to encompass and be generalisable across diverse populations (Rodriguez-Seijas et al., 2023). There are also discussions around connecting and collaborating research initiatives between HiTOP and the Research Domain Criteria (RDoC) to develop and establish bio-behavioural classifications of psychopathology (Michellini et al., 2021).

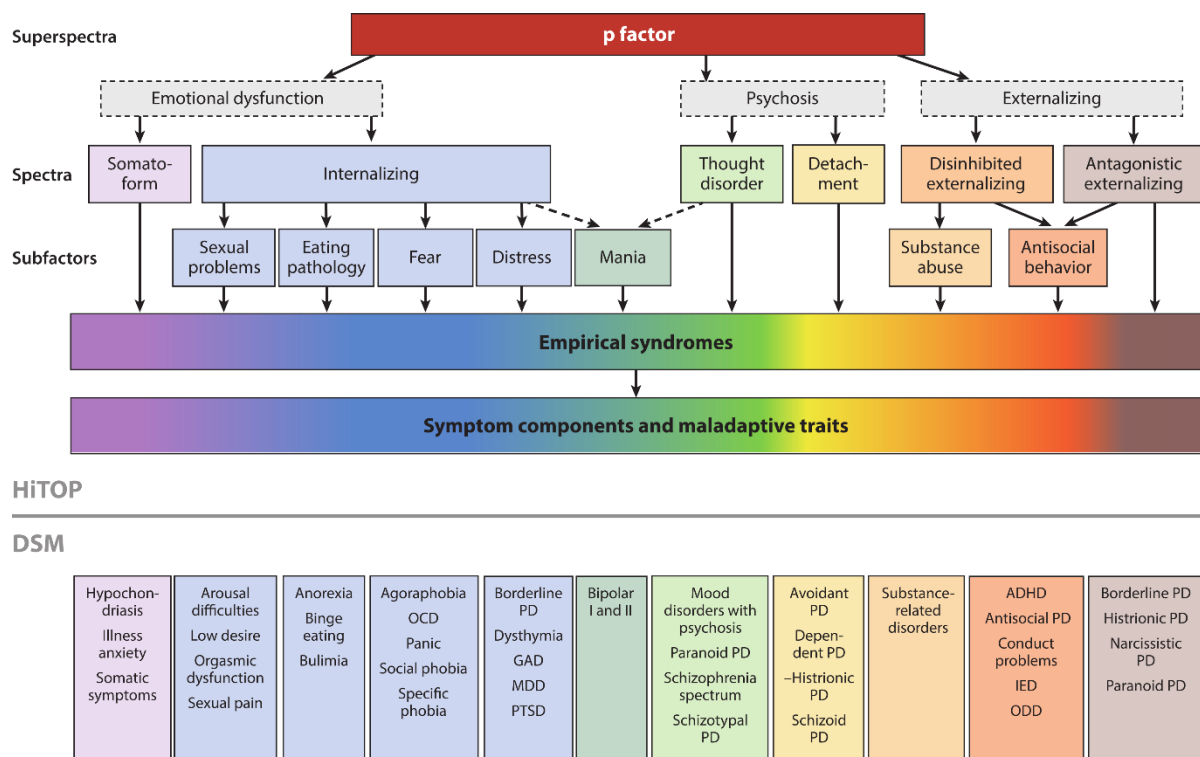


Figure 1: The HiTOP model (see Kotov et al., 2021). As mentioned by the authors, DSM diagnoses are not included in the model, but they are listed in accordance with corresponding symptoms that are clustered dimensionally.

As outlined by Kotov and colleagues (2021), and as seen in the figure 1, HiTOP is a hierarchical system, with levels of hierarchy corresponding to the specificity of symptoms,

with narrower, specific symptoms being hierarchically placed semantically underneath broader dimensions. At its broadest, HiTOP incorporates the p factor, representing an overarching risk for any psychopathology, with super-spectra, spectra, and specific problems represented as lower levels of the hierarchy, respectfully. These levels have been determined by data-driven insights, such as diagnostic comorbidities and factor analysis, whereby specific problems that are likely to co-occur would share a link to the same higher level of the hierarchy (Kotov et al., 2022). Ruggero and colleagues (2019) suggest that the framework allows clinicians to assess and collectively treat all symptoms and traits that encompass a patient's condition, regardless of their heterogeneity, and regardless of which diagnostic boundary they fall into, which is further explained with examples of therapeutic utility provided by Hopwood and colleagues (2020). While there is potential that this data-driven approach could be a step in the right direction for psychiatric nosology, there are unanswered questions around its comprehensiveness, utility, and the validity of a hierarchical structure. Tyrer (2018) questions the clinical utility that HiTOP would offer, questioning if the framework would improve treatment outcomes. This is a reasonable query, given that the proposed system does not yet address how current treatments for categorical diagnoses would align with dimensional ratings, and more importantly whether classifying conditions dimensionally would improve overall treatment or prevention outcomes, which is especially relevant since many patients seek identity, recognition, and even relief in what they consider to be a homogeneous condition (Saunders, 2023). However, as reported by Simms and colleagues (2022), much of the current research amongst the HiTOP consortium is aimed at addressing these issues, particularly on what a formal assessment measure would look like.

Shifting to dimensional diagnostic frameworks such as HiTOP would require evidence that such assessments would provide a more efficacious way of treating people, but for this to be effective the field still needs to understand how HiTOP would look in a clinical

setting, despite steps being made towards this by the HiTOP consortium (Ruggero et al., 2019). Possibly the most complicated challenge in establishing this is encompassing the dual relationship between commonality (i.e., the p factor) and discreteness (i.e., internalising and externalising factors). The p factor reflects the correlations between relatively narrower behavioural constructs, and while this represents that factors are related to each other, it also demonstrates the cumulative effect that aetiologies and symptoms together have of on the presentation of further symptoms. Understanding the nature of a given construct is vital to establish validity of such a model. Yet, with considerable literature now documenting the p factor, it is becoming abundantly clear that at the same time many risk factors can affect symptoms of a given dimension, and single risk factors can affect multiple different dimensions (Mamun et al., 2023; Lund et al., 2010; Harold & Sellers, 2018; Martin et al., 2018; Smoller et al., 2019; Middeldorp et al., 2016). Such understanding has prompted the need for life-course approaches which consider mental health across the lifespan and the trajectories of different symptom types (Caspi et al., 2020), to develop comprehensive clinical models which can assess a person's unique mental health pathway, rather than their current state at the time of clinical assessment.

HiTOP Gaps: Neurodevelopmental Symptoms

While HiTOP has aimed to comprehensively encompass symptoms corresponding with pre-existing diagnoses, and where they would fit along the broader spectra and sub-spectra, there are some symptoms for which the consortium feels there is not enough evidence to establish placement within the model, such as neurodevelopmental symptoms. For example, in the latest model, ADHD symptoms (i.e., inattention, hyperactivity, impulsivity) are conceptualised as being a narrower subgroup of symptoms represented as part of the broad externalising factor, mainly due to the co-occurrence with other externalising issues (Carragher et al., 2014), and symptoms related to autism spectrum

disorder (ASD) are absent from the model altogether (Conway et al., 2023). The exclusion of ASD symptoms from the model has been attributed to the lack of evidence regarding where they should be placed in the hierarchy (Kotov et al., 2021). The placement of ADHD symptoms under the externalising factor is driven by findings such as those of Carragher and colleagues (2014), which suggest that the inclusion of ADHD in the externalising spectrum results in the best fit being a continuous latent trait model, compared with a categorical latent class model. It is worth questioning the interpretation of these results though, given that the latent trait model only just had better fit than the latent class model, and both models were suggested to have good fit. Additionally, comparisons indicate that all models were very similar statistically in how they fit the data. The same study also conducted factor analysis to examine the structure of different externalising behaviours. Despite it loading onto the externalising factor, ADHD was by far the weakest indicator, indicating a shared and unique role in externalising issues and the structure of psychopathology more generally. Yet, due to such findings, ADHD symptoms have typically been lumped into the broad externalising dimension (Witkiewitz et al., 2013). Furthermore, while indeed ADHD symptoms are shown to be strongly associated with externalising symptoms across the lifespan (Armstrong et al., 2015; Kuja-Halkola et al., 2015; Carragher et al., 2014; Arcos-Burgos et al., 2012; Frodl, 2010; Lee et al., 2011; Obsuth et al., 2020), and typically highly comorbid with conduct disorder (CD) and Oppositional Defiant Disorder (ODD) throughout childhood (Harvey et al., 2016; Reale et al., 2017; Dellapiazza et al., 2021), there is a large amount of evidence suggesting that such symptoms can be statistically differentiated as being part of a broad neurodevelopmental factor (Lahey et al., 2018; Modabbernia et al., 2022; Stanton et al., 2021; Du Rietz et al., 2021; Miller et al., 2021; Michelini et al., 2019; Haltigan et al., 2018; Snyder et al., 2017; Bloemen et al., 2018). Additionally, a study by Du Rietz and colleagues (2021) indicates that despite ADHD symptoms being highly associated with externalising,

internalising, and other neurodevelopmental symptoms, after controlling for the general factor, only the relationship with other neurodevelopmental symptoms was sustained. These findings are further supported by reports indicating that ASD symptoms and ADHD symptoms have high rates of co-occurrence and share similar aetiologies (Hatch et al., 2023; Knott et al., 2021; Antshel et al., 2013; Miller et al., 2020; Antshel & Russo, 2019; Pettersson et al., 2013; Stergiakouli et al., 2017; Stanton et al., 2021), alongside other less prevalent diagnoses such as obsessive-compulsive disorder (OCD) (Kalin, 2020). Neurodevelopmental symptoms are also suggested to share transdiagnostic liabilities, such as social processing deficits (Baribeau et al., 2015), cognitive impairments (Dajani et al., 2016; Martin et al., 2015), and motor issues (Kramer et al., 2020).

Further evidence for the presence of a broad neurodevelopmental factor is represented by the largely shared genetic aetiology attributed to ADHD and ASD (Antshel et al., 2013; Cordova et al., 2022; Clarke & Eapen, 2021; Du Rietz et al., 2021), relative to externalising issues (Pettersson et al., 2013). Indeed, heritability estimates may be as high as 60-90% for ADHD (Waldman & Gizer, 2006), although more recent estimates place it at around 80% (Grimm et al., 2020). Similarly, heritability of ASD symptoms may be between 64-91% (Tick et al., 2016), although other findings place it more precisely at 85% (Yip et al., 2018). Other findings also suggest there may be aetiological specificity to neurodevelopmental symptoms. For instance, polygenic risk may predict more specificity in neurodevelopmental symptoms than in internalising and externalising symptoms (Loughnan et al., 2022). However, other work also clarifies the influence of the p factor, with polygenic scores for MDD being significantly associated with all five of the factors established by Michelini et al. (2019), and polygenic scores for ADHD being associated with the Externalising, Neurodevelopmental and Somatoform dimensions (Pat et al., 2022), indicating some uniqueness to their genetic aetiology, but also a large amount that is shared through the p

factor. These high genetic aetiologies are thought to reflect the onset of neurodevelopmental symptoms in early childhood (Pettersson et al., 2013; Harvey et al., 2016), and functional differences neurobiologically. Modabbernia and colleagues (2022) show measures of functional and structural connectivity to be related uniquely to neurodevelopmental symptoms, but not internalising or externalising symptoms. It is also worth noting, though, that in the same study measures of cortical surface area were found to be negatively associated with all dimensions, reflecting an association with the p factor, rather than specific dimensions, a finding which is line with work by Mewton and colleagues (2022). Thus, while there are some dimensions that may reflect more homogeneous samples, there is still likely a large amount of aetiological overlap between neurodevelopmental symptoms and other dimensions as well. Thus, there is already literature implicating the meaningfulness of a broad neurodevelopmental factor in psychopathology.

While statistical analyses may demonstrate that certain symptoms load better onto a given factor, hierarchical systems need to consider which correlations account for the most variance in a given dataset and determine dimensions of a hierarchy by meaningful distinctions. Thus, part of the issue with assessing shared and discrete associations is due to the difficulty accounting for the p factor. Neurodevelopmental symptoms have been shown to be highly associated with the p factor (Noordhof et al., 2015), which is also explained by high associations with both internalising (Karalunas et al., 2023; Du Rietz et al., 2021) and externalising issues (Carragher et al., 2014; Du Rietz et al., 2021). Indeed, a study by Noordhof and colleagues (2015) found that while ADHD and ASD symptoms both represented discrete factors from externalising and internalising issues, all factors were associated via the general factor. Also, as mentioned above, neurodevelopmental symptoms are highly associated with cognitive impairments, associations with are also shared with the general factor as well (Romer & Pizzagalli, 2021; Abramovitch et al., 2021; McGrath et al.,

2016). It is worth noting that despite the correlation of the p factor with cognition, other work assessing the association between cognition and mental health via early life adversity, found that, while both cognitive ability and mental health are affected by adversity, they were not specifically related to each other, indicating that, on top of shared aetiologies, there are likely unique additional factors which contribute to cognitive and mental health outcomes as well (Vedechkina & Holmes, 2023). This observation is supported by Chavez-Baldini et al. (2023), who show that while decreased cognitive functioning is associated with elevated psychopathology, they are independent of each other. Again, these findings support the hierarchical structure of HiTOP, due to differences in associations at different levels of specificity, but they allude the challenge of placing symptoms within a structural model based on correlations, rather than on aetiological findings.

Stability of Latent Factors

Despite the general basis of HiTOP having been shown to be equally applicable for adults and children (Kotov et al., 2022; Michelini et al., 2019), most factor analyses to inform such decisions (such as the study by Michelini et al. (2019)) have only analysed data from one time point. Insights derived from longitudinal findings have the potential to provide important details on the nature of the p factor and sub-factors across the lifespan, different aetiologies relevant to different domains and levels within the hierarchy, and then how we can address these issues clinically. For instance, HiTOP does not yet represent, consider, or explain the fluctuation of different symptom types across developmental stages, particularly the change in symptoms between childhood and adolescence, where more severe internalising issues tend to manifest (Blok et al., 2022; Caspi et al., 2014; Hankin et al., 2015). The gaps in the HiTOP framework are likely due to the scarcity of longitudinal studies which assess the stability and changes of dimensional constructs. Despite this, a large amount of the evidence used by HiTOP is related to the continuity of symptoms over time, whereby homotypic

continuity reflects the validity of a latent variable, and heterotypic continuity reflects the presence of the p factor (Lahey et al., 2021). Homotypic continuity refers to the relationship of a given symptom with itself at later timepoints, while heterotypic continuity refers to the association of a given symptom with other symptoms at later timepoints. Within homotypic continuity, symptoms can then be assessed as representing stability or change. More specifically, stability represents the consistency of symptoms across time, whereas change is associated with either decreases or increases in severity of that symptom. For instance, some symptoms with greater genetic liability may be more stable due to inherent biological effects, whereas symptoms with greater environmental liability may be more likely to fluctuate based on living circumstances and conditions. It is important to differentiate homotypic continuity from stability, as symptoms may have high levels of fluctuation but still be associated throughout different timepoints, whereas stability refers to the consistency of severity within homotypic continuity across these timepoints. For instance, all participants in a dataset could all increase by 1 score over each timepoint. This would reflect high homotypic continuity, but not stability.

Of the longitudinal literature that is available, homotypic and heterotypic continuity of most constructs are already well established, whereby it is already well documented that in general, symptoms in childhood are related to the further development of mental health problems in adolescence and adulthood (Groenman et al., 2017; Oldehinkel & Ormel, 2023; McElroy et al., 2018; Shevlin et al., 2017; Zablotsky et al., 2020; Newson et al., 2021; Fair et al., 2012), reflected in high levels of both homotypic and heterotypic continuity. In the Dunedin longitudinal study, Caspi and colleagues (2020) found that 70% of patients with an internalising disorder also received diagnosis of an externalising disorder at some point in time, and 77% vice versa, indicating a strong presence of the p factor, but also of potential heterotypic continuity between the internalising and externalising factors. To further affirm

this relationship, Winsper et al. (2020) demonstrate with parallel process latent class growth analysis that Internalising and Externalising symptoms have high rates of co-occurrence through childhood development (ages 4-11 in this study), although Internalising symptoms tended to gradually increase over this period, whereas Externalising symptoms remain consistent. Papachristou & Flouri (2020) ran a latent growth model on internalising symptoms, externalising symptoms, and cognitive ability, finding that all three types of symptoms appeared to have high rates of co-occurrence, and mostly shared developmental trajectory from ages 3 to 14. Blok and colleagues (2022) demonstrate findings with significant heterotypic continuity, showing in a latent transition analysis that those of any subgroup were 55% to 63% likely to cluster with a different subgroup at the next timepoint. Such subgroups were defined as being predominantly externalising, predominantly internalising, combined internalising and externalising, or normal functioning. Interestingly, while being one of the few studies to include ADHD symptoms in analysis, Shevlin and colleagues (2017) found ADHD symptoms to predict later externalising symptoms (but not vice-versa), findings which have also been established in other studies (Lee et al., 2011; Obsuth et al., 2020). These findings are further supported by earlier work by Lahey and colleagues (2014), who reported high rates of heterotypic continuity across diagnoses, even when controlling for homotypic continuity. The presence of homotypic continuity for individual sub-domains is also well established across psychopathology, with findings across internalising and externalising domains being associated with later symptoms in that their respective domains over time (Wichstrøm et al., 2017; Shevlin et al., 2017; McElroy et al., 2018; Snyder et al., 2017). What is not as well established is the stability of these symptoms over time. In other words, whether symptoms change or remain consistent regarding their severity over time. Additionally, most studies focus primarily on externalising and internalising as broad variables, without consideration of more specific factors (e.g.,

neurodevelopmental, somatoform, detachment) and the overarching p factor. Thus, longitudinal insights related to hierarchical factor structures are lacking.

While the general factor may demonstrate the cumulative effect of psychopathology as an interactome, longitudinal data may help us tease apart distinct trajectories, and more discrete and diagnosable aetiologies for sub-factors, as well as more specific symptoms. Much of the literature on psychopathology indicates that many symptoms related to severe internalising issues most often arise at some point throughout pre-adolescent development (i.e., ages 10-14), whereas externalising issues either plateau or decrease (Murray et al., 2020; Winsper et al., 2020; Blok et al., 2022), which has been suggested to be partly due to changes related to puberty and development (Nivard et al., 2017). The importance of this age bracket is not to be understated, as many of the behaviours developed here lay the foundation for adult mental health (Caspi et al., 2020). For further detail here, Blok and colleagues (2022) assessed CBCL ratings over three different time points. The first being between ages 5-8 (T1), then between ages 9-12 (T2), and finally at ages 13-16 (T3). Interestingly, there was an increase from 14.1% to 21% in psychopathological issues in general from T1 to T2, which then only increased to 22% in T3. Also, internalising issues increased in T1 to T2 from 5.1% to 9.2%, whereas externalising symptoms only increased from 7.3% to 8.3% from T1 to T2. Such findings indicate changes in severity of symptoms across these timepoints, representing greater changes. Additionally, as also suggested by Murray and colleagues (2020), these findings highlight the importance of changes between developmental stages (e.g., childhood and adolescence), as the change from childhood to adolescence (ages 10-13) results in increased internalising symptoms, as well as altered specificity of symptoms corresponding to certain dimensions. Contrastingly to the results of Blok and colleagues (2022), while studies using latent growth models are limited, findings from a paper by Hatoum and colleagues (2018b) suggest that both internalising and externalising symptoms are associated with

significant stability, by trajectories being better placed via the intercept (average score across timepoints) than the slope (change across timepoints). As is noticeable from the literature reviewed above, most of the longitudinal research on dimensional structures focus solely on broad internalising and externalising symptoms, except for Papachristou & Flouri (2020), who also included cognitive issues, and some other studies which have included ADHD (Wichstrøm et al., 2017; Obsuth et al., 2020; Shevlin et al., 2017). The analysis solely of broad internalising and externalising behaviours may limit the interpretation of such analyses, as factor models have been suggested to better fit the data when using a granular approach (Fernández de la Cruz et al., 2018).

As mentioned throughout this section, while the literature on continuity is substantial, studies assessing stability and change of broad dimensional latent constructs are lacking. Additionally, to our knowledge, no longitudinal studies have used latent factors determined from factor analysis, such as those identified in the study by Michelini and colleagues (2019). Such research is particularly needed for neurodevelopmental symptoms, for which placement within the HiTOP model remains in question (Conway et al., 2023). Despite this, it is already well established that neurodevelopmental symptoms typically emerge early in development and remain present over the life course to some extent (Miller et al., 2021). This suggests that the underlying causal mechanisms are likely due to factors which have long-lasting impact, reflected in latent growth curve models with what are called intercepts, rather than something that happens and changes the course of symptoms over time, which, in such models, are represented by slopes. Of research on neurodevelopmental symptoms, Wichstrøm and colleagues (2017) assessed homotypic and heterotypic continuity of diagnoses, indicating that ADHD had the highest rate of homotypic continuity, compared with ODD, CD, depression, and anxiety, although moderate homotypic continuity was also prevalent in such disorders. Such evidence points to the likelihood that a broad neurodevelopmental dimension will likely

represent similar homotypic continuity with dimensional data as well, when compared with broad internalising and externalising domains, which may indicate greater stability over time as well. It is also worth noting though, that such findings also indicate significant heterotypic continuity across all constructs (i.e., signifying the role of the p factor). Other research by Obsuth and colleagues (2020) supports such findings, with ADHD symptoms demonstrating the strongest estimates of homotypic continuity. Additionally, the same study produced findings indicating that ADHD symptoms are linked with externalising symptoms at later timepoints, but not vice-versa, demonstrating the early presence of ADHD symptoms likely related to early developmental and genetic causes, and the lack of links from other symptom types to ADHD symptoms. This is supported by research from Harvey and colleagues (2016), where findings of comorbidity between ADHD and ODD suggested that ADHD symptoms occur first and may play a role in the later development of ODD symptoms. This has also been observed with later internalising symptoms as well (Shevlin et al., 2017; Lee et al., 2011; Obsuth et al., 2020). These findings further warrant the consideration of a separate neurodevelopmental dimension in the HiTOP model, but no studies have yet assessed the stability of neurodevelopmental symptoms alongside other latent constructs longitudinally. Subsequently, due to the heterogeneity that is highly common with ADHD and ASD diagnosis (Antshel et al., 2013), as well as the association of ADHD symptoms with externalising issues (Carragher et al., 2014), further research is required to assess whether neurodevelopmental symptoms reflect more stable trajectories regarding severity, than externalising symptoms do, as this would further warrant distinction between neurodevelopmental symptoms and the externalising factor in hierarchical models such as HiTOP.

Birth Weight and Psychopathology

Determining the stability of neurodevelopmental symptoms is useful in establishing the nature of such symptoms longitudinally, but statistical differentiation only provides limited clinical insights without knowledge of aetiological factors that are driving such a distinction. As mentioned earlier, the genetic risk for ADHD and ASD symptoms is high relative to externalising and internalising symptoms (Pettersson et al., 2013; Cordova et al., 2022; Clarke & Eapen, 2021; Knott et al., 2021; Antshel et al., 2013), but other researchers hypothesise that neurobiologically, and functionally, neurodevelopmental symptoms can be further distinguished in factor models as a broad sub-domain (Stanton et al., 2021; Knott et al., 2021). However, as mentioned in these articles, further research is required to establish whether, for example, hyperactive and inattentive symptoms can be meaningfully differentiated from the broad externalising sub-factor in alignment with other neurodevelopmental issues such as repetitive and restrictive behaviours, stereotypical behaviour, and social impairment issues. An interesting relationship to investigate in this regard is that of psychopathology and birth weight. For instance, low birth weight (LBW) (< 2500g (5.51 lb)) is a measure that is also highly associated with multiple risk factors, such as preterm birth, maternal and paternal mental health, health problems, neurosensory impairment, household poverty, adverse childhood experiences, income, and prenatal exposure to tobacco or alcohol (Mamun et al., 2023; Rangan et al., 2020; Boyle et al., 2011; Momany et al., 2017), thus encompassing potential exposure to a range of aetiologies, and risk for psychopathology in general. However, even when controlling for the other risk factors mentioned above, lower birth weight has been found to be associated with elevated psychopathology in childhood and adolescence, and these symptoms are also likely to persist in pre-term and LBW individuals into adulthood as well (Linsell et al., 2019). Findings from a meta-analysis suggest that children born at low birth weight are 3.5 times more likely than

normal birth weight children to receive diagnosis for any mental disorder (Burnett et al., 2011). However, regarding the relationship of birth weight with dimensional factors, some have been more well documented than others, and in general, the findings indicate mixed results.

There is decent documentation on the relationship of birth weight with internalising symptoms (Boyle et al., 2011; Helle et al., 2019; Van Lieshout et al., 2018; Rangan et al., 2020), although there have been questions in such studies around the relationship with externalising symptoms, due to weak, small, or non-existent effects. Elsewhere there is some evidence for such a relationship though, due to findings from a meta-analysis of children aged between 5 and 12, by Mathewson and colleagues (2017), who assessed 41 studies which compared mental health reports of extremely low birth weight (ELBW) (< 1000g (2.2 lb)) with normal birth weight (NBW) (> 2500g (5.51lb)) controls. The findings of this study suggest that dimensionally rated externalising symptoms are in fact associated with ELBW (Mathewson et al., 2017), albeit with small effect sizes. As well as associations with dimensional ratings, this study also found associations of ELBW with diagnosis of CD, ODD, ADHD, and ASD. It is worth noting though, that the number of studies included in this meta-analysis that included analysis of externalising symptoms were quite low, so further research is needed to establish whether birth weight is involved in the development of externalising symptoms, as most of these associations also had small effect sizes. Additionally, there is no other evidence linking birth weight with externalising issues, making these results seem relatively questionable. Another meta-analysis provides further insights, in that lower birth weight was highly associated with diagnosis of any mental disorder (Anderson et al., 2021), although the relationship appears to be stronger between birth weight and neurodevelopmental issues. Diagnosis of ADHD was 18.8% for very low birth weight (VLBW) children, and 8.4% for those born at NBW. However, when breaking down

symptoms, there were considerable differences between different subtypes. 8.1% of VLBW (< 1500g (3.3lb)) were diagnosed as inattentive, compared to 4% of NBW. Contrastingly, only 3.4% of VLBW were diagnosed as hyperactive, compared to 2.9% of NBW. The most significant difference though was the combined type, which was 8.2% for VLBW, and only 2.9% for NBW, which may indicate that birth weight is associated with the p factor.

Diagnosis of ASD was also significantly higher for VLBW, at 5.6%, compared to only 0.5% of NBW. Anxiety and Mood disorders were also significantly associated with VLBW, although the relationship was weak for mood disorders. The researchers did not assess the relationship between birth weight and externalising issues.

In contrast, there is a further abundance of documentation on the relationship of birth weight with neurodevelopmental symptoms in childhood (Hatch et al., 2014; Yang et al., 2015). Lower birth weight has also been associated with increased ADHD symptoms (Pettersson et al., 2015; Ni et al., 2023) and diagnosis of ASD (Alhozyel et al., 2022). It is worth noting that the relationship between ADHD and birth weight has been shown to be partly mediated by neurocognitive ability (Morgan et al., 2018; Hatch et al., 2014) and genetic risk (Rahman et al., 2021). Yet, while there may be some mediators to this relationship, Ni and colleagues (2023) found that 14.4% of children with VLBW had diagnosis of ADHD, as did 11.5% of those with low birth weight, and 8.7% of those with normal birth weight, indicating that regardless of co-variables, there is an established spectral relationship between birth weight and ADHD symptoms, which is supported by findings presented by Dooley et al (2023b). Additionally, based on observations from twin studies, lower birth weight may be associated with ADHD symptoms due to nonshared environmental influences, rather than common genetic factors (Flicks et al., 2013). Such findings indicate that lower birth weight may act as a significant independent marker of psychopathology risk, particularly for the likes of ADHD. These findings can be further consolidated by a meta-

analysis which included 88 studies, which summarised that ADHD symptoms were significantly associated with lower birth weight (Momany et al., 2018). Additionally, a study using data from the ABCD study found that the attention problems subscale of the CBCL had by far the highest association with birth weight, compared to other subscales (e.g., rule-breaking behaviour, withdrawn-depressive symptoms) (Dooley et al., 2022). This study also found low birth weight to be associated with somatic symptoms, which has also been observed when assessing symptoms in adulthood (Berg et al., 2022; Westrupp et al., 2011). There have been (to our knowledge), no studies which have assessed the relationship between birth weight and detachment, although detachment can to some extent be thought of as a sub-factor of internalising symptoms (Michelini et al., 2019).

The ABCD Study and this Research

The Adolescent Brain Cognitive Development (ABCD) study is a multi-institution driven project, with a comprehensive array of measures being conducted across many different facets of health and well-being, such as psychiatric diagnoses, physical health, background details, neuroimaging, and many other measures of mental health and development (Karcher & Barch, 2021; Garavan et al., 2018). It is a longitudinal study, with children first beginning assessments at age 9/10, continuing annually and in some cases every six months throughout adolescence and into early adulthood. Thus, the ABCD study is set up ideally to assess longitudinal changes in behaviour, as well as the structural nature of psychopathology throughout these formative years. The study by Michelini and colleagues, as mentioned throughout several parts of this introduction, has provided a great starting point for understanding the factor structure of the CBCL (and psychopathology of children in general), the presence of the p factor in this model, and the validity between the child and adult models of HiTOP. However, such findings lack clarification needed from a confirmatory model, particularly which model (e.g., bifactor, hierarchical, correlated factors)

may best fit the data. Additionally, the continuity of these factors over time has yet to be assessed, particularly whether there are differences in stability between sub-factors. It is already well established that the p factor is very stable over time (Caspi et al., 2014; Romer & Pizzagalli, 2021; Caspi et al., 2020), and while there is evidence to suggest that the neurodevelopmental factor may be more stable over time (as mentioned earlier), to our knowledge there are no studies which have formally assessed this with a latent growth model.

Using data from the ABCD study, the aim of this study is to assess the stability of broad psychopathological subdomains (internalising, externalising, neurodevelopmental, somatoform, detachment) determined by Michelini and colleagues (2019), and the trajectories of development for these factors across time, in hope of adding to longitudinal findings to literature that can aid the development of the HiTOP model. Our first hypothesis is related to the change of symptom severity over time. We expect that, because of chronic effects due to the reportedly higher genetic/heritable component of neurodevelopmental symptoms, relative to internalising and externalising symptoms (Du Rietz et al., 2021), the neurodevelopmental factor will have greater stability and less change, relative to the other subdomains. We have no expectations regarding the nature of the slope for the general factor, although evidence suggests that p factor scores may be somewhat stable over time (Caspi et al., 2014; Romer & Pizzagalli, 2021; Caspi et al., 2020). The stability of the neurodevelopmental factor and other factors will be represented through latent growth models, of which we expect the neurodevelopmental factor to provide better fit when included as an intercept, and worse fit than other sub-domains when included as a slope. Although we expect the general factor to be associated with the intercept, we also expect it to be associated with the slope as well. Despite internalising and externalising symptoms being predominantly associated with the intercept in previous research (Hatoum et al., 2018b), we expect neurodevelopmental symptoms to have greater associations with the intercept. While not a direct hypothesis, there

is decent evidence to suggest that specification of a slope for internalising and externalising factors will improve fit in the latent growth models, as per Dekovic and colleagues (2004), whereas neurodevelopmental symptoms will not.

Our second hypothesis is centred around the relationship of psychopathology with birth weight. Due to previous research indicating that lower birth weight is associated with increased risk for internalising symptoms and neurodevelopmental symptoms, we expect birth weight to also be associated with the p factor, and thus, be associated with each of the factors as well. In this sample, Dooley and colleagues (2022) already found attention problems to be most associated with lower birth weight, using the original subscales of the CBCL. These results lack applicability to hierarchical models though, due to findings suggesting the original CBCL subscales lack validity compared with factor driven clusters (Brislin et al., 2022). Also, because the neurodevelopmental factor established by Michelini and colleagues (2019) is a broader collection of symptoms, it is worth considering that birth weight correlations may be somewhat different to those found by Dooley and colleagues (2022), although their findings are mainly in line with our hypothesis. Interestingly, despite withdrawn depressed and anxious depressed subscales not being associated with birth weight in the study by Dooley and colleagues (2022), other research indicates the presence of a relationship between internalising symptoms and lower birth (Boyle et al., 2011; Helle et al., 2019; Van Lieshout et al., 2018; Rangan et al., 2020). Again, there is potential for the differences between the preset CBCL subscales (e.g., withdrawn depressed, anxious depressed) to produce different results to the subscale identified by Michelini and colleagues (2019) (e.g., internalising). Thus, our hypothesis here is based on the collection of previous findings in other samples, whereby we expect elevated internalising symptoms will be associated with lower birth weight. Despite evidence for a relationship between lower birth weight and somatic symptoms in adulthood (Berg et al., 2022; Westrupp et al., 2011), to our

knowledge the only evidence of somatic symptoms in childhood being associated with lower birth weight comes from the moderate effect found by Dooley and colleagues (2022). Thus, analysis of the relationship between birth weight and somatic symptoms will be exploratory, as it also will for externalising and detachment symptoms, given there is no current evidence for a relationship here. More specifically, our hypothesis is that neurodevelopmental, internalising, and the general factors will be associated with birth weight via their intercept in the latent growth model, rather than the slope, for which we expect there to be no relationship. This is due to the slope representing deviation of a given individual away from the average, which is more likely due to present environmental factors influencing the child, rather than neonatal changes which happened before birth. Due to reported sex differences in CBCL answers (van der Sluis et al., 2016), we will analyse all results by assessing differences in males and females. Results from this study could help inform researchers and mental health clinicians on the presentation and nature of symptom continuity across different dimensional spectrums of psychopathology. Such understandings could then be used or incorporated in the HiTOP model (or other models), particularly for neurodevelopmental symptoms, for which placement is still uncertain.

Method

ABCD Study

The Adolescent Brain Cognitive Development (ABCD) study is a multi-institution driven project, including nearly 12,000 participants and a parent or caregiver, with data being gathered longitudinally from ages 9 or 10 until aged 20. Data includes a vastly wide range of research methods and measures, such as clinical diagnoses, health information, background details, family details, neuroimaging, mental health questionnaires, genetics, alongside many others (Karcher & Barch, 2021; Garavan et al., 2018). Currently the study is up to release 5.0, which includes data from baseline and four annual follow-ups, although some measures have

been captured bi-annually every 6 months. Because our research was conducted outside of the United States and the universities that run the ABCD study, approval from the United States National Institution of Mental Health (NIMH) National Data Archive (NDA) was necessary to gain access to the data. This access is available to researchers of affiliated external organisations with federal wide assurance from the NIMH. Questionnaires from the ABCD study which were included in this study were the child behaviour checklist (CBCL), a demographics survey, a developmental history questionnaire (DHQ), as well as an administrative questionnaire. Items not required for analysis were removed from each dataset in the cleaning process. Each questionnaire is explained in more detail below.

Child Behaviour Checklist

The Childhood Behaviour Checklist (CBCL) is a questionnaire that consists of 119 questions related to child mental health and behaviour (Achenbach & Rescorla, 2001). Each question is rated out of 0, 1, and 2 in relation to the presence of the symptom. 0 being '*not true*', 1 being '*somewhat/sometimes true*', and 2 being '*very/often true*'. Questions can be broadly classified as being part of 8 subscales, namely anxious depressed, withdrawn depressed, somatic problems, social problems, thought problems, attention problems, rule breaking, and aggressive behaviour. However, a factor analysis conducted by Michelini and colleagues (2019) suggests that the CBCL is better characterised as having 5 subscales, namely internalising, externalising, somatoform, neurodevelopmental, and detachment. Each of these subscales were then correlated with a general factor, although alternative factor structures of this model have not yet been assessed. As these subscales have been empirically derived, we chose to use them for our own subsequent analyses.

Developmental History Questionnaire

The DHQ is an ABCD questionnaire related to developmental information, which is answered retrospectively by the participant's parent, caregiver, or guardian (Dooley et al.,

2023a). The NIMH data dictionary provides detailed descriptions of each variable in the questionnaire (see here: https://nda.nih.gov/data_structure.html?short_name=dhx01). We used 11 variables specific to the DHQ, including birth weight measures in pounds and ounces, mother's age at birth, father's age at birth, does the child have a twin, was the child born prematurely, and if so, how many weeks premature was the child born. Retrospective accounts for information such as birth weight has been shown to have strong validity (Peterson et al., 2019; Kassem et al., 2018). Some cleaning was done to the raw data to prepare it for analysis. Those who answered 'do not know' for any of the questions were re-coded as NA for analysis purposes. Answers for birth weight which were below the current recorded lowest and heaviest birth weights were re-coded as NA due to implausibility. Birth weight was measured by parent or caregiver recall in pounds and ounces. 514 participants did not provide an answer in pounds, while 1,979 did not provide an answer for ounces. We classed those who did not provide an answer in pounds as being missing and rounded those who had provided an answer in pounds but not in ounces to 0. We opted to re-calculate birth weight so that it was measured in grams. This process required additionally re-coding two answers as 0, as these answers recorded values for ounces beyond the threshold of 16 and did not equal the other answer in pounds. Because these two answers still provided reasonable answers for birth weight in pounds, we opted to leave that part of their answer as it was. To re-calculate birth weight, we multiplied ounces by 6.25 so each value was decimally equivalent to 100 (as per the metric system). We then multiplied the combined pounds/ounces (to 100) value by 453.592, before rounding each number to 0 so it was a single measurement in grams. In grams, this resulted in a minimum birth weight value of 821, and a maximum of 6690.

Demographics

The demographics questionnaire is an ABCD questionnaire related to demographic information about a participant. Answers were provided by the child's parent, caregiver, or guardian. The NIMH data dictionary provides detailed descriptions of each variable in the questionnaire (see here: https://nda.nih.gov/data_structure.html?short_name=pdem02). We used 19 questions for our study, including the language spoken by the respondent (only English or Spanish were available options), the respondent's relationship to the child (Options: biological mother, biological father, adoptive parent, custodial parent, or other), the child's age, the child's sex at birth, the child's gender identity, the caregiver's age, the caregiver's gender, the caregiver's education level, the caregiver's income, the household's total income, the number of people currently living in the household, the child's ethnicity, as well questions related to poverty (e.g., 'In the past 12 months, has there been a time when you and your immediate family needed food but couldn't afford to buy it or couldn't afford to go out to get it'). Two participants answered 60 and 77 for the number of people living in their household, which was deemed unlikely so both answers were re-coded as NA. For any answers where the respondent had selected 'do not know' or 'refuse to answer', we re-coded as NA.

Participants

Participants are from the ABCD dataset of 11,868 United States children, aged between 8 and 11. At baseline, the sample consists of 0.2% 8-year-olds, 52% 9-year-olds, 47.6% 10-year-olds, and 0.2% 11-year-olds. Participants complete follow-up tasks, questionnaires, etc. every year following baseline collection, which after four years places them between 12 to 15 years old. Interview age was also recorded in months, which ranged from the youngest at 107 (8 years 11 months) to the oldest at 133 (11 years 1 month). A detailed description of the recruitment of the ABCD study has been published by Garavan

and colleagues (2018). The sample includes 6188 (52.14%) males, 5677 females (47.83%), and 3 intersex males (0.03%). For any analysis where we assessed sex differences, the intersex participants were removed from the data. Most children were reported as identifying with either the male or female genders, with 4 children identifying as ‘different’, another 4 identifying as ‘trans female’, 2 identifying as ‘gender queer’, and another 2 identifying as ‘trans male’. 9 values were missing for child gender. Ethnic backgrounds of children were 52% White, 20% Hispanic, 15% Black, 11% Other, and 2% Asian, which are considered to be representative of the United States population in general (Garavan et al., 2018). 2 Responses for race ethnicity were already coded as missing (NA). 651 Respondents answered in Spanish, the rest all answered in English. The caregiver respondents were primarily biological mothers (85.3%), followed by biological father’s (10%), adoptive parent’s (2.3%), other (1.4%), and custodial parent’s (1%). Previous work by Paul and colleagues (2021) who also used the CBCL demonstrated that results remain the same even when excluding parent or caregiver respondents who were not the mother, so we opted to include all respondents. 89% of parents/caregivers identified as a woman, while 10.8% identified as a man. 9 caregivers (0.08%) identified as trans male, 6 (0.05%) identified as gender queer, 5 (0.04%) identified as trans female, and 4 (0.03%) identified as different. 7 answers were missing from parent/caregiver gender identity. The average age of respondents was 40, with the youngest being 23, and the oldest being 80. 91 values were missing for caregiver age.

Data Cleaning

After gaining access to the ABCD data from the NIMH NDA database, downloaded files were stored securely on the University password protected OneDrive system, and have been deleted following analysis. We first cleaned the data by re-coding values that provided no information (e.g., ‘refused to answer’ or ‘do not know’) as missing values (NA). As already established by Michelini and Colleagues (2019), some CBCL items already share

high correlations, so we averaged these values together to create composite items, as per their paper. For factor analysis, these averaged values had to be rounded to 0 decimal places, so they were able to be measured as ordinal. The developmental History Questionnaires consisted of some answers which were from baseline and some which were from later timepoints. We opted to only include baseline data, as this was the only timepoint which provided comprehensive responses from each participant.

Our analysis consisted of two phases: Firstly, the process of determining factor structure, and secondly, the process of assessing stability and change amongst said structures over time. For the first phase of our analysis, we were only interested in CBCL values, so included all participants whose data was collected for the CBCL. Sample size decreased over time due to missing values for participants who either had not yet completed follow-up responses or had been removed from the study. Subsequently, this meant there were 11,861 participants at baseline, 11,201 participants for the first annual follow up, 10,897 participants at the second annual follow up, 10,098 participants at the third annual follow up, and 4,678 participants at the fourth annual follow up. We chose not to include the fifth wave as only 4,678 participants had submitted responses for this timepoint, which is a substantial drop from the other timepoints. Our second phase of analysis had different sample sizes due to the inclusion of birth weight as a variable. Because twins are considered to have lower birth weight than single born children, due systematic gestational differences (Dooley et al., 2022), we chose to remove the 2224 twins from the sample. For this question, 37 responses were recorded as ‘don’t know’, and data was missing for 3 participants, so these were also re-coded as missing (NA) and subsequently excluded from analyses. Prior to removing twins and missing values from the dataset there were 11,868 participants included, twin removal resulted in a dataset which included 9,604 participants. For dealing with missing data in all

calculations, we used full information maximum likelihood (FIML) which has been suggested to be good at accounting for missing data (Lim & Cheung, 2022)

For our birth weight regressions, we included some covariates to control for socioeconomic and pre-term factors that have also been shown to be associated with psychopathology, these were combined household income, caregiver education level, and whether birth was considered as pre-term. Caregiver education and combined household income were re-coded in line with another study in the ABCD cohort (see Paul et al. (2021) for further details). Due to term of birth only measured as either pre-term or term, we opted to include the categorical measure of pre-term birth, as opposed to a variable which asked how many weeks pre-term the child was born. This decision was determined based on the low number of reports of children born pre-term, thus, categorising pre-term birth creates a much more even distribution.

Confirmatory Factor Analysis (CFA)

For our preliminary analyses we tested the factor structure of the CBCL across four time points. Alternative factor models were estimated at each time point with sub-domains based on factor analysis by Michelini and colleagues (2019), who used data from CBCL scores in the ABCD study to establish a five-factor model as being the best solution for the data. This study only employed exploratory factor analysis (EFA), without comparing alternative models. Thus, we compared alternative factor models using the *lavaan* package in R (Rosseel, 2012). The five factors established in the study by Michelini and colleagues (2019) had already been labelled as Internalising, Externalising, Neurodevelopmental, Somatoform, and Detachment. Model fit can be determined by comparing RMSEA and CFI values between each model (West et al., 2012). We compared the fit of a base model (each item only loading onto its own sub-domain), hierarchical model (each item loads onto its own sub-domain, and each sub-domain loads onto a broad general factor), and bi-factor model

(each item loads onto its own sub-domain, as well as the general factor). Results suggested that the bi-factor model best fit the data, further detail can be found in the results section, and in Appendix A.

Expanding on CFA results, we decided to exploratorily assess bi-factor indices, to see which factors accounted for most of the variance amongst the data. We report on the findings of ECV_SS, ECV_GS, ECV_SG, Omega, OmegaH, the Omega Ratio, FD, and H. Explained Common Variance (ECV), which can be used to evaluate the uni-dimensionality of common variance in a factor structure (Rodriguez et al., 2016), is reported in three ways (Dueber, 2017). Firstly, ECV_SS represents ECV of a subscale factor itself. ECV_GS represents the ECV of the general factor accounted for by each specific factor, and ECV_SG represents the ECV of each specific factor accounted for by the general factor. Factor Determinacy (FD) is the correlation between factor scores and factors, with higher scores reflecting better determinacy (Rodriguez et al., 2016). Construct Replicability (H) reflects the level to which a set of items represent a latent variable. Values greater than 0.8 are suggested to represent a stable latent construct (Rodriguez et al., 2016). Omega is a model-based estimate that represents the proportion of variance in the unit-weighted total score that is attributed to all sources of common variance, whereas Omega Hierarchical represents the estimated proportion of variance attributed to each single factor that is attributable to the general factor (Rodriguez et al., 2016). As per Rodriguez et al. (2016), high Omega values indicate reliability of the latent construct and the values underlying it. These values are typically associated with the number of items loading to each factor, whereby latent variables with more items are likely to have higher Omega values. Omega Hierarchical also tends to increase with the number of items loaded onto a factor, but it is highly influenced by the loadings of each item on the either the general factor or its associated subscale. Subsequently, higher Omega Hierarchical scores tend to represent higher influence of the general factor on

the estimated variance, whereby scores above 0.8 are considered unidimensional, due to most of the variance being attributed to the general factor. The Omega Ratio (mentioned as omegaR from here onwards) is the proportion of reliable variance attributed to a given factor, which is calculated by dividing omegaH over omega (as demonstrated by Hatch et al., 2021).

Measurement Invariance

While analysis by Michelini and colleagues (2019) assessed the model separately in males and females, to our knowledge, formal assessment of measurement invariance has not yet been conducted for this factor structure, so we wanted to check that the latent factor of focus reflected the same construct over time and between sex. Previous research has established measurement invariance of sex for the general factor (Lahey et al., 2018), and on the preset CBCL subscales (van der Sluis et al., 2016), but such findings did not consider narrower latent constructs. Previous work within the ABCD sample found the five subscales established by Michelini and colleagues (2019) to be longitudinally invariant over three timepoints (Romer & Pizzagalli, 2021), but these analyses were only conducted on a base model, thereby measurement invariance on this factor structure in a bi-factor model has not been established yet. With the fourth wave available as well, we decided to assess the model with all subscales included as a bi-factor model, and over each of the four waves available from the ABCD release 5.0.

We assessed longitudinal invariance by creating factor models which compared data between each timepoint, with CBCL items as ordinal variables, as per Liu et al. (2017). For each of these timepoint comparisons, we compared four models with different constraints: Firstly, configural invariance (no constraints), secondly, metric invariance (loadings constrained), thirdly, a model where loadings and thresholds were constrained, and lastly, scalar invariance (loadings, thresholds, and intercepts constrained) (Putnik & Bornstein, 2016), using the CFA framework from previous analyses, as per Meredith (1993), with theta

parameterisation (Wu & Estabrook, 2016). Invariance is determined by using a chi-square difference test, whereby a model is considered not invariant by observing a difference of no more than 0.01 for CFI, and 0.015 for RMSEA (Cheung & Rensvold, 2002). Because there was no difference at all between the model where thresholds were constrained with the metric invariance model (at any comparison of timepoints), we did not report these values (as they are all duplicate values of metric invariance). Results suggested that this factor structure was longitudinally invariant, as is demonstrated by the outputs reported in the results and Appendix B.

For groupwise measurement invariance of sex, we compared four different models. The first model was a configural model where thresholds were constrained to be equal across groups. The second model, which was compared with the configural model was a metric invariance model (where thresholds and loadings were constrained). The third model, which was then compared to the metric invariance model was a model where thresholds, loadings, and intercepts were constrained. The last model constrained thresholds, loadings, intercepts, and residuals. This model was then compared with the previous model, where intercepts had been added as being constrained. The same fit criteria apply as is explained in the previous paragraph, the main difference being the groupwise comparison of males and females, rather than the within-sample comparison of the same individual at different timepoints. No studies to date have assessed measurement invariance for sex on the factor structure established by Michelini and colleagues (2019), although several studies have reported sex differences amongst different factors of the CBCL (Dekker et al., 2007; Gross-Tsur et al., 2006; Aitken et al., 2019), so it is worth checking to be sure that the model is consistent for both males and females, as would be good practice even if sex differences had not been reported. Results suggested that indeed this factor structure is measurement invariant for sex. These outputs are briefly reported in the results section, and can be seen in more detail in Appendix C.

Latent Growth Curve Modelling

For our main analyses we tested the stability and change in the general and sub-domains of the bi-factor model and the relationship of each factor with birthweight, by creating latent growth models using the *lavaan* (Rosseel, 2012), and *semTools* packages in R (Jorgensen et al., 2016). Due to the CBCL consisting of a large number of items, and some of the sub-domains consisting of a vastly different numbers of items (Michellini et al., 2019), a model including raw items would have been too complex and likely hard to interpret (Newsom, 2024). Because we had established measurement invariance of the bi-factor structure across time and sex, we parcelled subsets of items within each sub-domain to create broad composite items, in order to create a less complex latent growth model. It was important to consider measurement invariance before parcelling, given the biased effect this can have on measurement invariance (Lee & Whittaker, 2021). Parcelling allows model estimation to be simplified by reducing the number of items so there are fewer indicators (Yang et al., 2010), creating a smaller model, which has been suggested to improve model fit (Rhemtulla, 2016). Parcelling was done by using the *psych* package in R (Revelle, 2024) with the functions *polychoric* and *iClust*, the latter of which assesses how different items cluster together (Revelle, 1979). We used the *iClust* function to identify item clusters based on polychoric correlations of items loading onto a sub-domain. For this process specifications are made so that 3 parcel items are created for each sub-domain (e.g., Internalising_1, Internalising_2, Internalising_3, Externalising 1, and so on), except for detachment, which only had 5 items loaded onto it (Michellini et al., 2019). Hence, this sub-domain could only be parcelled into two broader variables (e.g., if it was parcelled into three items, one of the parcels would still be one of the original items, as the items are averaged together). Thus, we ended up with 14 parcel items from the five sub-domains. To create each parcelled variable, we calculated the average of each item that loaded onto it. Where one item loaded onto

multiple composites, we included it in the composite score for that which it loaded highest on to. Item parcel loadings can be seen in Appendix D.

Latent growth modelling is a type of structural equation modelling especially used for analysing individual differences that are stable or change over time (Newsom, 2024). A model is established by specifying an intercept (reflecting individual differences in underlying latent construct that are stable over time) and slope (reflecting individual differences in a latent construct that increases or decreases over time). Comparisons of model fit when including either an intercept or a slope to the model can provide insights as to whether the data is better represented by stability or change (Newsom, 2024). Because our hypothesis is questioning whether symptom severity changes across time for different sub-domains, we are interested in whether such changes can be accounted for by an underlying construct that is specific or general, which can be assessed by analysing the fit differences between a model with solely intercepts included, and a model where the slope of a given factor is specified. Chi-squared difference tests can be used to compare fit between such models (Pavlov et al., 2020). We used the parcelled items to estimate models accounting for change in the general and sub-domain factors, using the bi-factor model established by CFA in prior analysis. We first estimated a model, using the *sem* function as part of the *semTools* package (Jorgensen et al., 2016), where the general and sub-domain factors loaded onto second order factors that capture stable individual differences in the general and specific sub-domain factors over time. Thus, this model included only the intercepts, and a slope for the general factor, representing a model where scores of each factor are stable over time, but where the general factor captures change in symptom severity. Intercepts and slopes are specified as second order factors that account for either stability or change of symptom severity in first order latent factors, respectively. Intercepts are specified by constraining symptom severity to remain consistent across each timepoint (e.g., $G_i \approx 1 \cdot G_A + 1 \cdot G_B$

+ 1*G_C + 1*G_D). In the example provided, G_i represents the intercept for the general factor, and G_A, G_B, etc represent the general factor score at such timepoints. Second, we tested in turn models that accounted for linear change in the general factor and linear change in either of the sub-domain factors. This was done by creating several additional models, each of which represented the linear change of a given factor (i.e., six models: One each with a slope for the general, externalising, internalising, neurodevelopmental, somatoform, and detachment factors). Change was specified by parameterisation of the slopes that indicate linear increases (e.g., $G_s \sim 0 \cdot G_A + 0.5 \cdot G_B + 1 \cdot G_C + 1.5 \cdot G_D$). The example provided here demonstrates for the general factor that symptoms are expected to increase steadily at each time point. For each model constraints were made so that each factor could not correlate together, since shared value is already captured by the general factor, as per Snyder et al., (2017). Latent growth model parameters were specified as follows: meanstructure = TRUE, Estimator = MLR (maximum likelihood estimation with robust (Huber-White) standard errors and a scaled test statistic), missing = FIML). Once the models were specified, we compared fit using chi squared difference tests. We compared the fit of the model that specified stable individual differences across time (no slopes) with each model where change was specified for a given factor (slope included for one factor at a time). We then compared the fit differences between model comparisons to assess which factors were best represented by a model with just the intercept, and which model was best represented by the inclusion of slope parameters. As per our hypothesis, we expected that the neurodevelopmental factor would have less fit difference in the chi-squared difference test than other factors, due to expected lack of data alignment with the slope specifications.

Using the latent growth models with slopes specified, we also tested if birth weight was associated with either stable or linear changes in individual differences in the general and/or sub-domains. In previous research birth weight has been categorically coded as being

either extremely low birth weight (ELBW) (Mathewson et al., 2017), very low birth weight (VLBW) (Anderson et al., 2021), low birth weight (LBW) (Mamun et al., 2023), or normal birth weight (NBW) (see section on birth weight in the introduction for further details).

However, this approach fails to account for the continuous nature of birth weight, whereby there is always potential for a given correlation to be positively associated. Thus, we decided not to categorically code birth weight. We assessed the relationship of birth weight with each factor by specifying it to be a predictor of each slope and intercept one model at a time. This would result in 12 models (i.e., six factors, one intercept and slope for each factor), each with a regression identifying birth weight as a predictor of the intercept or slope. Because we aimed to assess the regression of birth weight with 12 different models, we took a conservative approach to the p values (Jafari & Ansari-Pour, 2019), by considering anything below 0.004 as significant (i.e., $0.05/12 = 0.004$). Because there are also multiple known risk factors established as being associated with psychopathology (Paul et al., 2021), we included three variables as covariates and present regressions of birth weight with and without covariates. Covariates included combined household income (here on also referred to as family income), caregiver education level, and pre-term/premature birth. For regressions of birth weight, we divided values in grams by 1000 so variance was not disproportionate to latent variable variance. We also report the standardised effects (std.lv), which reflects the amount of standard deviation change associated with one unit change of a predictor and indicates the direction of the relationship.

Results

Preliminary Analysis

We used Comparative Fit Index (CFI) and Root Mean Square Error of Approximation (RMSEA) to assess model fit for CFA. CFI of > 0.95 and RMSEA of < 0.05 are considered to indicate a good model fit (West et al., 2012). Results for these analyses can be found in

Appendix A. All models showed good fit regarding RMSEA and CFI values (West et al., 2012). The standard factor model for this solution had an RMSEA of 0.034, the hierarchical model had an RMSEA of 0.035, and the bi-factor model had an RMSEA of 0.029, indicating that the bi-factor model best fit the data. While there has been extensive debate around the best choice of model between correlated factors, hierarchical, and bi-factor solutions (Greene et al., 2019; Bonifay & Cai, 2017; Waldman et al., 2023), with much of the research suggesting bi-factor models may be biased (Greene et al., 2019). However, previous work from Clark and colleagues (2021) in this dataset indicated that the theoretical differentiation between bi-factor and hierarchical solutions is minimal, so we subsequently chose to run with the bi-factor model, due to it having a significantly better model fit. This also replicates other previous work within this dataset (Moore et al., 2020), and other studies which assessed the factor structure of the CBCL (McElroy et al., 2018; Haltigan et al., 2018).

Findings of bi-factor indices are also reported in detail in Appendix A. The outputs show consistency across each timepoint. While each latent variable had high Omega values, factors with a higher number of items loaded to it had higher Omega values. This is a particularly important consideration for our research, given that there is a large gap between each factor regarding the number of items loaded onto it. For instance, the detachment subscale only has five items underlying it, whereas the externalising factor has 36 (although some of these items were made to be composite items). At baseline, the Omega value (rounded to 2 decimal places) was 0.92 for the Internalising factor, 0.93 for the neurodevelopmental factor, 0.97 for the externalising factor, 0.86 for the somatoform factor, 0.85 for the detachment factor, and 0.98 for the general factor. Interestingly, while OmegaH values have been reported to also be influenced by the number of items (Rodriguez et al., 2016), there were some significant differences between each subscale. At baseline, the OmegaH value (rounded to 2 decimal places) was 0.17 for the Neurodevelopmental factor,

0.32 for the Internalising factor, 0.31 for the Externalising factor, 0.53 for the Somatoform factor, 0.35 for the Detachment factor, and 0.87 for the general factor. These values remained reasonably consistent over time, although there were decreases for both the neurodevelopmental and somatoform values. Thus, with OmegaR being OmegaH divided by Omega (Hatch et al., 2021), at baseline, the OmegaR (rounded to 2 decimal places) was 0.35 for the Internalising factor, 0.18 for the neurodevelopmental factor, 0.32 for the externalising factor, 0.62 for the somatoform factor, 0.41 for the Detachment factor, and 0.89 for the general factor. These findings indicate that the general factor accounted for by far the most amount of reliable variance, relative to each of the specific factors, whereas the neurodevelopmental factor accounted for the least amount of reliable variance.

As mentioned briefly in the methodology, the bi-factor model was confirmed to be longitudinally invariant across each timepoint. RMSEA values remained consistent across configural, metric, and scalar models (see Appendix B for detailed results). Because the model which constrained loadings and thresholds provided no change in fit whatsoever, we opted to only report on configural, metric, and scalar invariance. The bi-factor structure was also found to be invariant for groupwise sex differences as well, and this was consistent when assessing this at each timepoint (see Appendix C for detailed results).

Latent Growth Model Differences

Results from chi squared difference tests indicated that for all sub-domains adding a slope (e.g., $G_s \sim 0 \cdot G_A + 0.5 \cdot G_B + 1 \cdot G_C + 1.5 \cdot G_D$) to the model improved fit indices, suggesting that the data better fits the inclusion of a slope, albeit these relationships were small for some factors. The somatoform factor showed the least change in symptom severity (i.e., least difference between models), with a X^2 difference of 26.042. The neurodevelopmental factor had the next least changes, with an X^2 difference of 54.709. Chi squared differences for other factors can be seen below in Table 2. We also report sex

differences between models, which can be seen in Table 3 and Table 4. Interestingly, notable differences between males and females were apparent. For instance, male X^2 differences across Externalising (28.533), Internalising (28.477), Somatoform (20.198), and Neurodevelopmental (32.105) factors all suggested minimal change, although the inclusion of a slope still appeared to create a slightly better fit of the data. Whereas for female X^2 differences for the Internalising (76.077) and Externalising (79.621) factors remained just as pronounced, while differences for the Somatoform (9.31) and Neurodevelopmental (24.467) factors were much lower. Possibly the most striking difference is that the Detachment factor had an X^2 difference of 429.01 when including a slope to the model. This effect was evident for both males and females, although it was more pronounced in females. Below in table 1 we show the standardised estimates for second order slopes respective of each latent factor. These values show the extent to which slopes differ from the intercept.

Table 1: *Standard estimates and p values for second order slopes*

All Participants	P	std.lv (SE)
General Slope	< 0.001	-0.413 (0.002)
Externalising Slope	0.013	0.096 (0.001)
Internalising Slope	< 0.001	0.295 (0.003)
Neurodevelopmental Slope	< 0.001	0.367 (0.002)
Somatoform Slope	0.001	0.136 (0.002)
Detachment Slope	< 0.001	0.857 (0.008)
Females		
General Slope	< 0.001	-0.453 (0.003)
Externalising Slope	0.007	0.14 (0.002)
Internalising Slope	< 0.001	0.573 (0.005)
Neurodevelopmental Slope	< 0.001	0.475 (0.004)
Somatoform Slope	< 0.001	0.321 (0.003)
Detachment Slope	< 0.001	0.971 (0.008)
Males		
General Slope	< 0.001	-0.434 (0.003)
Externalising Slope	0.16	0.092 (0.002)
Internalising Slope	0.263	0.071 (0.003)
Neurodevelopmental Slope	0.001	0.422 (0.004)
Somatoform Slope	0.494	0.034 (0.002)
Detachment Slope	< 0.001	0.777 (0.005)

Notes: P values of less than 0.05 are considered statistically significant, these are highlighted in bold. Std.lv represents the standardised estimates, and standard error (SE) is in brackets. Standardised estimates represent the amount of change in standard deviation across time. Presented here are estimates for males and females, as well as when all participants are included in the model.

Table 2: *Chi squared difference test of latent growth curve models*

All Participants	df	AIC	BIC	X ²	X ² Difference	P(>X ²)
Comparison Model	1438	-233325	-231791	12536		
Externalising Slope	1435	-233555	-231999	12300	84.921	< 2.2e-16
Internalising Slope	1435	-233445	-231889	12411	74.353	4.99E-16
Somatoform Slope	1435	-233364	-231808	12492	26.042	9.35E-06
Neuro Slope	1435	-233412	-231856	12444	54.709	7.92E-12
Detachment Slope	1436	-233961	-232412	11896	429.01	< 2.2e-16

Table 3: *Chi squared difference test of latent growth curve models for females*

Sex: Females	df	AIC	BIC	X ²	X ² Difference	P(>X ²)
Comparison Model	1438	-118177	-116802	7812.8		
Externalising Slope	1435	-118400	-117006	7583.6	79.621	< 2.2e-16
Internalising Slope	1435	-118295	-116901	7688.5	76.077	< 2.2e-16
Somatoform Slope	1435	-118186	-116792	7797.5	9.31	0.02544
Neuro Slope	1435	-118214	-116820	7769.6	24.467	2.00E-05
Detachment Slope	1436	-118592	-117204	7393.8	290.82	< 2.2e-16

Table 4: *Chi squared difference test of latent growth curve models for males*

Sex: Males	df	AIC	BIC	X ²	X ² Difference	P(>X ²)
Comparison Model	1438	-119973	-118576	6572.7		
Externalising Slope	1435	-120037	-118621	6502.5	28.533	2.81E-06
Internalising Slope	1435	-120016	-118600	6523.9	28.477	2.89E-06
Somatoform Slope	1435	-120002	-118586	6537.9	20.198	0.000154
Neuro Slope	1435	-120021	-118604	6519.2	32.105	4.97E-07
Detachment Slope	1436	-120195	-118786	6346.6	154.48	< 2.2e-16

Birth Weight Regressions

Due to the neurodevelopmental slope resulting in errors when estimated, we ran 11 complete models, although we include the neurodevelopmental model in consideration of p values, given that it was involved in repeated significance testing. We did this for one model at a time, with each model specifying birth weight as being a predictor of a given intercept or

slope, one latent variable at a time. Thus, we assessed significant p values as being 0.004 or less ($0.05/12 = 0.004$). Before controlling for covariate associations, as hypothesised, we found *lower* birth weight to be significantly associated with the neurodevelopmental intercept, but only for boys. Surprisingly, we also found that *higher* birth weight was significantly associated with the internalising intercept for both boys and girls, despite previous findings suggesting that *lower* birth weight is associated with internalising issues (Boyle et al., 2011; Helle et al., 2019; Van Lieshout et al., 2018; Rangan et al., 2020). However, after controlling for covariates, such as caregiver education level, combined household income, and pre-term birth, no latent variables were significantly associated with birth weight in either direction. P values, as well as standardised estimates (std.lv) and standard errors (SE) can be seen in tables 4 and 5 below. Table 4 shows stand-alone regressions of psychopathology intercepts and slopes with birth weight, table 5 represents the same regressions when also controlling for covariates. In each table we separately report associations for males and females.

Table 5: *Latent growth regressions of birth weight (BW) with the intercept and slope of each factor*

BW Regression	All Participants		Females		Males	
Intercepts	P	std.lv (SE)	P	std.lv (SE)	P	std.lv (SE)
General	0.014	-0.052 (0.003)	0.186	-0.042 (0.004)	0.011	-0.073 (0.004)
Neuro	0.035	-0.054 (0.003)	0.281	-0.04 (0.004)	0.001	-0.116 (0.004)
Internalising	< 0.001	0.122 (0.003)	0.002	0.139 (0.004)	0.001	0.128 (0.003)
Externalising	0.521	-0.014 (0.002)	NA	NA	0.051	-0.056 (0.003)
Detachment	0.429	-0.024 (0.004)	0.081	-0.073 (0.006)	0.795	0.011 (0.006)
Somatoform	0.152	-0.039 (0.002)	0.902	-0.005 (0.004)	0.683	-0.015 (0.003)
Slopes						
General	0.259	0.043 (0.002)	0.059	0.109 (0.002)	0.468	0.038 (0.002)
Neuro	NA	NA	0.373	0.256 (0.002)	NA	NA
Internalising	0.123	0.072 (0.002)	0.033	0.146 (0.003)	0.443	0.047 (0.002)
Externalising	0.694	0.021 (0.001)	0.674	-0.027 (0.002)	0.984	0.002 (0.002)
Detachment	0.002	-0.14 (0.004)	0.008	-0.161 (0.006)	0.178	-0.093 (0.005)
Somatoform	0.254	0.103 (0.002)	0.022	0.394 (0.003)	0.787	0.029 (0.002)

Notes: P values represent the probability that the predictive variable (birth weight) is correlated with psychopathology domains. Because we ran these regressions for twelve different models, we took a conservative approach to assessing p values ($0.05/12 = 0.04$), therefore any values below 0.004 were considered statistically significant. Std.lv represents the standardised estimates, indicating the direction of the relationship and to what effect, while standard error (SE) (in brackets) represents the standard error of these values. Negative values in the std.lv column indicate a negative relationship between birth weight and psychopathology (i.e., lower birth weight predicted higher levels of symptoms). Due to the poor fit of the neurodevelopmental slope, standardised estimates of that regression model resulted in errors. We chose to report the p values for the neurodevelopmental slope regardless, despite no values being significant. The errors are likely due to the data not fitting a neurodevelopmental slope very well. Also note that the neurodevelopmental factor is shortened to 'neuro' here.

Table 6: *Latent growth regressions of birth weight (BW) with psychopathology, controlling for covariates*

BW Regression	All Participants		Females		Males	
Intercepts	P	std.lv (SE)	P	std.lv (SE)	P	std.lv (SE)
General	0.669	0.01 (0.003)	0.382	0.031 (0.004)	0.5	-0.022 (0.004)
Neuro	0.757	-0.009 (0.003)	0.654	-0.019 (0.004)	0.066	-0.072 (0.005)
Internalising	0.027	0.072 (0.003)	0.026	0.108 (0.005)	0.055	0.078 (0.004)
Externalising	0.363	0.022 (0.002)	NA	NA	0.288	-0.033 (0.003)
Detachment	0.891	0.005 (0.005)	0.171	-0.068 (0.007)	0.307	0.05 (0.007)
Somatoform	0.261	-0.036 (0.003)	0.697	0.019 (0.004)	0.743	-0.014 (0.003)
Slopes						
General	0.382	0.036 (0.002)	0.101	0.1 (0.003)	0.515	0.038 (0.002)
Neuro	NA	NA	NA	NA	NA	NA
Internalising	0.658	-0.023 (0.002)	0.26	0.083 (0.004)	0.454	-0.051 (0.002)
Externalising	0.411	0.049 (0.001)	0.854	-0.014 (0.002)	0.736	0.031 (0.002)
Detachment	0.216	-0.063 (0.004)	0.282	-0.072 (0.007)	0.913	-0.009 (0.005)
Somatoform	0.663	0.043 (0.002)	0.057	0.281 (0.003)	0.748	-0.044 (0.003)

Notes: As per table 4, P values represent the probability that the predictive variable (birth weight) is correlated with psychopathology domains. Because we ran these regressions for twelve different models, we took a conservative approach to assessing p values ($0.05/12 = 0.04$), therefore any values below 0.004 were considered statistically significant. Std.lv represents the standardised estimates, indicating the direction of the relationship and to what effect, while standard error (SE) (in brackets) represents the standard error of these values. Negative values in the std.lv column indicate a negative relationship between birth weight and psychopathology (i.e., lower birth weight predicted higher levels of symptoms). Due to the poor fit of the neurodevelopmental slope, standardised estimates of that regression model resulted in errors. We chose to report the p values for the neurodevelopmental slope regardless, despite no values being significant. The errors are likely due to the data not fitting a neurodevelopmental slope very well. Also note that the neurodevelopmental factor is shortened to 'neuro' here. Also, regression of the externalising intercept in females resulted errors, likely due to the extremely low mean values and lack of variance for females in externalising issues.

Discussion

Interpretation of Results: Slope Models

As per our first hypothesis, the neurodevelopmental factor was associated with relatively less change than the internalising and externalising factors, although the difference was less than anticipated, and results should be considered cautiously, given that standardised estimates for the neurodevelopmental slope placed it as being significant (i.e., there are some changes in neurodevelopmental symptoms for this sample over time), sex differences across each sub-domain were noticeable, and differences between model fit were not seen in standard estimates of the neurodevelopmental slope. This suggests that neurodevelopmental symptoms may be more liable to change over time than previously hypothesised. It is worth noting that a study by Kuja-Halkola and colleagues (2015) also produced findings suggesting that neurodevelopmental symptoms may to some extent be liable to change over time. These changes may in part reflect the potential for neurodevelopmental symptoms to be affected by a liability to both developmental deficits (e.g., birth weight and premature birth), which predict stable symptom severity from neo-natal stages onwards, and environmental liabilities (e.g., poverty), which affect change in symptom severity at later timepoints. However, results are somewhat mixed, given that fitting regressions to the slope of the neurodevelopmental factor resulted in errors, which is likely due to a neurodevelopmental slope not being a good representation of the data (i.e., low variance from the intercept). Despite the neurodevelopmental factor being hypothesised to represent the least change across time, the somatoform factor represented the least amount of change, with very little difference in fit between a model with the somatoform slope included and the comparison model, which was sustained for males and females. Interestingly, the detachment factor by far showed the least stability across time, due to the addition of a detachment slope creating largely greater fit than the comparison model, which indicates that this age range (9-13) may be a key age for the

development of these symptoms. It is worthwhile to note that there were also substantial sex differences between model comparisons. While somatoform and neurodevelopmental factors showed little difference between respective slope models and comparison models, this stability appeared to be reflected for both males and females. Yet, externalising and internalising symptoms showed moderate stability relative to other factors when including all participants but showed far greater stability for males than females. Additionally, while the detachment slope model still produced a largely better model fit than the comparison model, this difference was far more pronounced for females. Such findings indicate different developmental trajectories for males and females, whereby pre-adolescence may reflect far greater change for female symptoms than males.

Interpretation of Results: Birth Weight Regressions

Our second hypothesis was that we expected the neurodevelopmental factor, the internalising factor, the somatoform factor, and the general factor to be associated with lower birth weight via the intercept from latent growth models, but not the slope. After controlling for covariates, we found neither the general factor, nor any of the sub-domains to be associated with lower birth weight via their respective intercept or slope models. Before including covariates, the neurodevelopmental intercept was associated with lower birth weight, but only for boys, matching findings by Dooley and colleagues (2022), who also used the ABCD dataset. Surprisingly, before including covariates, higher birth weight was significantly associated with the internalising intercept. This points to the importance of measuring birth weight continuously, as many previous studies have only assessed birth weight categorically, eliminating the potential to analyse the relationship between elevated birth weight and psychopathology. Interestingly, while birth weight was not significant for any factors after controlling for covariates, pre-term birth was highly associated with the general factor intercept and the neurodevelopmental intercept, which is surprising, given the

strong relationship between birth weight and premature birth (Anderson et al., 2021; Yang et al., 2015). Other researchers have suggested that it is not birth weight itself that is associated with psychopathology, but the effects that lead to low birth weight (e.g., neo-natal issues or complications, poor maternal health) (Ji et al., 2024), which are perhaps better measures of functional development, and may explain the lack of associations between birth weight and psychopathology after controlling for covariates. Regardless, it appears that the lack of latent factors being associated with birth weight is due to the covariates (e.g., combined family combine) having a greater role in the development of psychopathology. This indicates that associations between birth weight and psychopathology found in other studies may be due to it having a shared relationship with other risk factors, as mentioned in the introduction. Interestingly, despite high associations of birth weight with pre-term birth, pre-term birth appeared to have a stronger relationship with neurodevelopmental symptoms, as well as that of the general factor. Our findings aligned with other work in the ABCD dataset, where pre-term birth was associated with general psychopathology (Ji et al., 2024), although analysis here provides further details as to what factors of psychopathology are more/less associated with pre-term birth. Surprisingly, despite pre-term birth predicting higher scores on the general factor for both males and females, the neurodevelopmental factor was the only sub-domain to be significantly associated with pre-term birth. Thus, it may be worth considering that pre-term birth, rather than birth weight, more accurately reflects developmental differences that are thought to be a strong predictor of stable neurodevelopmental symptoms over time (Ji et al., 2024).

Covariate Regressions

While they were primarily used to control for external risk factors, other covariates also shed light onto the nature of each sub-domain, these results are discussed exploratorily, and can be seen in Appendix E. The strongest covariate by far appeared to be combined

household income, which was significantly associated with elevated symptoms for intercepts of the general factor, the externalising factor, and the detachment factor. Surprisingly, higher income predicted elevated internalising symptoms for the intercept and slope, an association that may explain why higher birth weight was associated with the internalising intercept before controlling for covariates. Furthermore, higher caregiver education was significantly associated in predicting elevated internalising symptoms in females for both the slope and the intercept, but not males. Thus, it appears there may be a trend for caregivers of females with higher socioeconomic status to rate their child higher on internalising symptoms than those with low socioeconomic status. Contrastingly, elevated neurodevelopmental symptoms were associated with lower caregiver education for the intercept, but not for the slope.

Additionally, this relationship was only sustained for females, not males, indicating that onset of neurodevelopmental symptoms may be affected by aetiological differences respective of sex. Other work within the ABCD study has indicated that low socioeconomic status has a much stronger relationship with psychosocial problems than pre-term birth (de Laat et al., 2016), as also found by Santos and colleagues (2017). Given that pre-term birth and birth weight are highly correlated (Anderson et al., 2021; Yang et al., 2015), this provides further explanation as to the lack of relationship between birth weight and psychopathologies when including covariates, due to the very strong relationship between income/socioeconomic status and general psychopathology. It is worthwhile to note that the neurodevelopmental factor still showed unique associations with predictors, due to it being the only factor associated with pre-term birth. Additionally, it is worth noting that the neurodevelopmental and somatoform factors, which showed the least difference to the comparison model when slopes were specified, were the only two factors not to be associated with family income. This indicates that even though the neurodevelopmental symptoms may be subject to some changes in severity across time, aetiological factors are more likely to be due to biological or

neo-natal disruptions that happen early in life, which may provide further basis for distinction from the externalising factor.

Exploratory Insights

While not of primary focus regarding our hypotheses, the exploratory assessment of bi-factor indices provides insight into the nature of this factor structure and childhood psychopathology sub-domains. For instance, as can be seen in Appendix A, the neurodevelopmental factor had the lowest value of omegaH at baseline (0.17) (these values remained consistent across each timepoint), relative to all other factors, indicating that variance within the neurodevelopmental factor was largely unaccounted for by the general factor, and that instead this latent construct represents the multidimensionality of this factor structure. Indeed, higher scores of omegaH reflect a greater proportion of variance explained by the general factor (Rodriguez et al., 2016), suggesting that relative to the other latent constructs in this factor structure, the neurodevelopmental factor is the most unique. It is worth noting that the internalising (0.32), externalising (0.31), and detachment (0.35) factors also still had relatively low omegaH scores at baseline, while the somatoform factor had highest omegaH score (0.53), indicating that the model generally explains a mixture of multi-dimensionality and uni-dimensionality among more specific factors, which further consolidates evidence that psychopathology may be most accurately described as a hierarchical structure of symptoms (Kotov et al., 2021). This also demonstrates that for the somatoform factor, which had higher omegaH values at each timepoint than the other factors, more of its variance is accounted for by the p factor (Rodriguez et al., 2016). Additionally, the general factor by far had the greatest omegaH value, and because this was above 0.8, it can be considered as a reliable uni-dimensional construct (Rodriguez et al., 2016).

Applying Findings to Hierarchical Models

As per a multitude of studies now (Martel et al., 2017; McElroy et al., 2018; Laceulle et al., 2015; Brislin et al., 2022; Romer et al., 2018; Castellanos-Ryan et al., 2016; Lahey et al., 2018; Patalay et al., 2015; Caspi et al., 2014; Romer & Pizzagalli, 2021; Caspi et al., 2020), the effect of the general factor (p factor) is deeply established as being a real, and prominent feature in the factor structure of psychopathology. While delineation here of granular sub-domains is statistically well-founded, it is evident that the p factor still has a significant role in driving symptom severity across sub-domains, likely due to the evidence suggesting that many aetiologies effect a wide range of symptoms. As seen by the bi-factor indices derived from our CFA, the general factor accounts for by far the largest amount of variance between sub-domains and the data. Yet, these findings implicate the importance of considering the presence of both multidimensionality and uni-dimensionality as being the defining characteristics of hierarchical models, especially in respect to aetiologies which are relevant to specific and broad symptoms respectively. While it is becoming increasingly established that neurodevelopmental symptoms differentiate from externalising symptoms in factor analysis (Micheline et al., 2019; Stanton et al., 2021), due to several of our findings, there is evidence to implicate why neurodevelopmental symptoms load onto to their own factor and can subsequently be meaningfully differentiated from these factors. As mentioned earlier, neurodevelopmental and somatoform symptoms showed less change over time (i.e., low X^2 differences with comparison model) than the externalising, internalising, and detachment factors. These differences are further highlighted by the errors present when trying to fit regressions to the neurodevelopmental slope, as well as regressions themselves, whereby the neurodevelopmental intercept was greatly associated with pre-term birth, which was not the case for any other sub-domain. Additionally, neurodevelopmental symptoms were not significantly associated with family income at either the intercept or the slope.

Contrastingly, lower family income was associated with both the externalising intercept and slope, indicating that lower socioeconomic status may not only affect baseline externalising symptoms, but can also increase the likelihood that these symptoms increase in severity over time, or vice versa, where improving socioeconomic status may help decrease the severity of these symptoms over time. Furthermore, the neurodevelopmental intercept was significantly associated with lower caregiver education, a finding which was not found for the externalising factor, or any other factor. Overall, these findings suggest that while the neurodevelopmental factor still loads onto a general psychopathology factor, there is a strong element of uniqueness reflected in both aetiology and trajectory.

Despite evidence in this study suggesting that there is a lack of change regarding symptom severity of the neurodevelopmental factor across time, it is still unknown whether differentiating neurodevelopmental symptoms in a diagnostic framework such as HiTOP would provide more efficacious treatment outcomes. This is particularly relevant given the high rates of comorbid symptoms between dimensions, and findings from Acarturk and colleagues (2019) which show that treatments which are target symptoms in general, rather than specific symptoms, are most effective at treating patients with comorbid symptoms such as depression and anxiety. Furthermore, while neurodevelopmental symptoms largely co-occur, they can still be substantially different in severity and can present heterogeneously (Antshel & Russo, 2019). In general, this is an area of literature which requires substantially more research before determining how these observations can be implemented into clinical practice.

Limitations and future considerations

While insights gained from the findings of this study are valuable, they come with several limitations and worthwhile considerations regarding the interpretation of the results. Firstly, labelled as the neurodevelopmental factor by Michelini and colleagues (2019), the

generalisability of this term should be considered somewhat cautiously, given the lack of ASD symptoms represented in the CBCL. For instance, there are two factors related to ASD: (1) Social communication, and (2) Repetitive/Restrictive Behaviours (Kamp-Becker et al., 2009; Shuster et al., 2014), which are typically measured with a separate questionnaire to the CBCL. The only question which assesses restrictive, repetitive behaviours in the CBCL is Question 66 (Repeats certain acts over and over; compulsions). Thus, because questions that assess the social component of ASD are not measurable with the CBCL, our findings cannot fully extend to ASD related symptoms that are considered part of the neurodevelopmental spectrum (Kamp-Becker et al., 2009; Shuster et al., 2014). Therefore, while these findings may provide distinction of attention and hyperactivity symptoms within hierarchical models, it is beyond the scope of this study to suggest placement of the neurodevelopmental factor within the HiTOP model, given how closely associated ASD symptoms are with attention and hyperactivity symptoms (Hatch et al., 2023; Knott et al., 2021; Antshel et al., 2013; Miller et al., 2020; Antshel & Russo, 2019; Pettersson et al., 2013; Stergiakouli et al., 2017; Stanton et al., 2021). Other studies, such as a paper by Bloemen and colleagues (2018), have used the Child Social Behaviour Questionnaire (CSBQ) to accommodate for this gap, a questionnaire which is aligned with the CBCL's three-point scale, making it easily comparable.

Unfortunately, the CSBQ is not measured in the ABCD study, and including other measures was deemed as unfeasible. The differentiation of ADHD and ASD beneath the potential neurodevelopmental super-spectra is important, due to the overall uniqueness of ASD (Knott et al., 2021). Interestingly, findings by Noordhof and colleagues (2015) indicated ASD and ADHD symptoms may both represent their own unique factors, which may also explain higher rates of environmental risk attributed to ADHD (Kunitoki et al., 2022; Wüstner et al., 2019) than for ASD (Tick et al., 2016), although further analysis may be required here, to determine whether this distinction remains after accounting for the p factor. Distinctively, it

is worth noting that adult onset of ADHD is thought to be associated with much lower heritability estimates (around 35%) (Brikell et al., 2019), indicating that while symptoms of inattention and hyperactivity may have a strong genetic liability for childhood onset, they may also share liability with environmental factors too. Furthermore, it has been suggested that a meaningful distinction can be made between children with ADHD that co-occurs with more frequently with externalising behaviours, and ADHD which co-occurs with ASD (Dellapiazza et al., 2021), due to differences in social impairment. Additionally, genes implicated in the onset of ADHD symptoms are thought to be distinct from those involved at later stages, whereby some genes are involved in the initial onset of symptoms, and some are related to developmental changes, resulting in heterogeneous samples (Brikell et al., 2019). Yet, despite high rates of heritability in ASD, the presentation of genetic abnormalities is highly heterogeneous. This is important to consider, given the hundreds of genes which are implicated (Vortsman et al., 2017), and the potential for any of the genetic mutations involved to be (or not be) present for a given person, which creates a seemingly infinite number of unique presentations. Furthermore, and in line with such findings, symptoms of ASD are highly heterogeneous too (Kamp-Becker et al., 2009). Thus, future work aimed at determining the presence of a neurodevelopmental factor within the hierarchical structure of psychopathology should consider the issues listed above carefully.

While these findings have been centred on the concept of change across time with respect of each sub-domain, the observation of either stability or change in symptom severity is more useful when paired with an understanding of what is driving the stability or change in said symptom severity. We included birth weight (as well as covariates) as an independent variable in our analyses, but the findings suggested that the covariates used have much more influence over psychopathology symptoms than birth weight. Future research should investigate a comprehensive range of risk factors as potential aetiologies to help further

develop understanding of these latent constructs, especially regarding what drives stability or change in their severity, and what drives the differences in sub-domains beneath the general factor. It is worth considering that while the p factor represents commonality of psychopathology symptoms, there is still an observed differentiation between sub-domains, suggesting that there are both unique and shared aetiologies across different types of symptoms in each sub-domain.

While the ABCD study provides a large dataset that enables enormous potential for statistical analysis, the study has its own limitations which are important to mention. For instance, in most childhood studies, parents tend to rate symptoms as higher than the child themselves or their teacher (Boyle et al., 2011; Helle et al., 2019), although it is unclear whether this is due to biases, inaccuracy, or other factors. Additionally, symptoms related to developmental information (e.g., birth weight and pre-term birth) were measured retrospectively, meaning that accounts of these variables may not always be objective. Despite indications that retrospective accounts of these measures offer decent reliability (Petersen et al., 2019), they should still be considered with caution due to a lack of objectivity. Subsequently, it is considered best practice to record scores from multiple informants, which may be a worthwhile area of focus for future longitudinal studies.

Conclusions

Findings from our study indicate that neurodevelopmental symptoms may be represented by unique aetiology and function relative to other latent factors (e.g., internalising, externalising) due to differences in associations with pre-term birth, family income, and caregiver education. These findings are supported by lower fit values in models that include a slope for the neurodevelopmental and somatoform factors, relevant to other domains. Moving forward with dimensional models, it will be important to be able to differentiate different developmental trajectories across different dimensions, to tailor

treatment solutions to the right people. While chi-squared differences between models which specified slopes for each latent factor were minimal, there were some indications that the neurodevelopmental factor represents less changes throughout time than the internalising and externalising factors. While no factors were associated with birth weight when controlling for covariates, the findings above are further supported by associations of the neurodevelopmental intercept with pre-term birth, and a lack of association with family income. Additionally, it is worthwhile to add that the detachment factor by far captured the greatest amount of change across time, highlighting this age group as a potential risk period for such symptoms to become elevated. Our findings also supported what is now an abundance of literature in support of the p factor as an overarching dimension of a hierarchical structure for psychopathology (Kotov et al., 2021), as demonstrated by it accounting for by far the largest amount of variance in bi-factor indices. Additionally, family income appears to be the strongest predictor of psychopathology in general relative to neonatal measures. Future research should look at further timepoints beyond ages 13/14 to see how trajectories develop throughout later adolescence. It would also be useful for such research to include measures of ASD symptoms, as this may help to better substantiate the presence of a broad neurodevelopmental factor, or more specific factors related to ADHD and ASD issues. Overall, these findings signify that a neurodevelopmental factor can be differentiated from other sub-domains, particularly the externalising factor. Such findings may help inform further research and development within the HiTOP model, and research in psychiatric nosology in general.

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Appendices

Appendix A

Below are the results of each CFA for the five-factor model established by Michelini and colleagues (2019). As per the table below, the bi-factor model best fit the data, although all models showed good fit (West et al., 2012). As mentioned, we also assessed bi-factor indices from the bi-factor model that showed the best fit. All values are rounded to 2 decimal places, these outputs can be seen below the CFA Results table.

Table 7: *Fit Indices from CFA analyses*

CFA Results	RMSEA	CFI	TLI	X ²	DF	SRMR
Standard	0.034	0.977	0.976	41651.293	2904	0.064
Hierarchical	0.035	0.975	0.974	44691.27	2909	0.067
Bi-Factor	0.029	0.982	0.981	31840.335	2836	0.056

Note: CFA results from standard, hierarchical and bi-factor models. Root mean square error of approximation (RMSEA) values below 0.05 are suggested to indicate good fit. Comparative fit index (CFI) values above 0.95 are suggested to indicate good fit, as are Tucker Lewis Index (TLI). X² represents the chi squared test value. DF represents degrees of freedom. Standardized root mean square residual (SRMR) represents the difference between the residuals of a sample correlation matrix and a hypothesised correlation matrix. See Schumacker & Lomax (2010) for more detailed information on fit indices.

Table 8: *Bi-Factor Indices of the factor model at the first timepoint measured (T1)*

T1	ECVss	ECVsg	ECVgs	Omega	OmegaH	OmegaR	H	FD
Int_A	0.37	0.06	0.63	0.92	0.32	0.35	0.74	0.88
Neuro_A	0.21	0.05	0.79	0.93	0.17	0.18	0.66	0.82
Ext_A	0.34	0.14	0.66	0.97	0.31	0.32	0.87	0.91
Som_A	0.64	0.07	0.36	0.86	0.53	0.62	0.81	0.92
Det_A	0.42	0.03	0.58	0.85	0.35	0.41	0.62	0.86
GA	0.64	0.64	0.64	0.98	0.87	0.89	0.97	0.97

Notes: Each factor is abbreviated (Internalising = Int, Neurodevelopmental = Neuro, Externalising = Ext, Somatoform = Som, Detachment = Det, General = G), and has a letter after it to reference which timepoint the table represents (T1 = A, T2 = B, T3 = C, T4 = D). Details related to the descriptions of each value can be found in the methodology section.

Table 9: *Bi-Factor Indices of the factor model at the second timepoint (First follow-up)*

T2	ECVss	ECVsg	ECVgs	Omega	OmegaH	OmegaR	H	FD
Int_B	0.39	0.07	0.61	0.92	0.34	0.37	0.75	0.89
Neuro_B	0.21	0.05	0.79	0.93	0.17	0.18	0.66	0.82
Ext_B	0.34	0.14	0.66	0.97	0.31	0.32	0.87	0.92
Som_B	0.66	0.08	0.34	0.86	0.55	0.63	0.83	0.93

Det_B	0.42	0.03	0.58	0.84	0.35	0.41	0.62	0.88
GB	0.64	0.64	0.64	0.98	0.86	0.88	0.97	0.97

Notes: Each factor is abbreviated (Internalising = Int, Neurodevelopmental = Neuro, Externalising = Ext, Somatoform = Som, Detachment = Det, General = G), and has a letter after it to reference which timepoint the table represents (T1 = A, T2 = B, T3 = C, T4 = D). Details related to the descriptions of each value can be found in the methodology section.

Table 10: *Bi-Factor Indices of the factor model at the third timepoint (Second follow-up)*

T3	ECVss	ECVsg	ECVgs	Omega	OmegaH	OmegaR	H	FD
Int_C	0.40	0.07	0.60	0.92	0.36	0.38	0.77	0.89
Neuro_C	0.19	0.04	0.81	0.93	0.15	0.16	0.63	0.81
Ext_C	0.34	0.14	0.66	0.97	0.31	0.32	0.87	0.91
Som_C	0.61	0.07	0.39	0.87	0.51	0.59	0.81	0.93
Det_C	0.43	0.03	0.57	0.86	0.36	0.42	0.64	0.88
GC	0.64	0.64	0.64	0.98	0.87	0.89	0.97	0.97

Notes: Each factor is abbreviated (Internalising = Int, Neurodevelopmental = Neuro, Externalising = Ext, Somatoform = Som, Detachment = Det, General = G), and has a letter after it to reference which timepoint the table represents (T1 = A, T2 = B, T3 = C, T4 = D). Details related to the descriptions of each value can be found in the methodology section.

Table 11: *Bi-Factor Indices of the factor model at the third timepoint (Second follow-up)*

T4	ECVss	ECVsg	ECVgs	Omega	OmegaH	OmegaR	H	FD
Int_D	0.38	0.07	0.62	0.93	0.34	0.37	0.76	0.89
Neuro_D	0.17	0.04	0.83	0.93	0.13	0.14	0.60	0.80
Ext_D	0.36	0.15	0.64	0.97	0.34	0.35	0.88	0.92
Som_D	0.58	0.07	0.42	0.88	0.49	0.56	0.80	0.92
Det_D	0.44	0.04	0.56	0.86	0.36	0.42	0.66	0.90
GD	0.64	0.64	0.64	0.98	0.87	0.88	0.97	0.97

Notes: Each factor is abbreviated (Internalising = Int, Neurodevelopmental = Neuro, Externalising = Ext, Somatoform = Som, Detachment = Det, General = G), and has a letter after it to reference which timepoint the table represents (T1 = A, T2 = B, T3 = C, T4 = D). Details related to the descriptions of each value can be found in the methodology section.

Appendix B

Below are the results of longitudinal measurement invariance. As can be seen for each comparison of timepoints, RMSEA values, CFI values, and TLI values all remained consistent, indicating invariance between each timepoint. The configural model represents no constraints on any of the parameters, so acts as a comparator when constraints are made. The metric model represents loadings being constrained to be equal between each timepoint, thus if each loading was exactly equal across each timepoint the model fit would be the same as the configural model. We also tested a model where loadings and thresholds were

constrained, but since these fit indices were the same as the metric invariance fit outputs, we opted to not include them in these tables. Scalar invariance then represents a model where loadings, thresholds, and intercepts are all constrained to remain equal across each timepoint.

Table 12: *Measurement Invariance between baseline (T1) measures and 1-year follow-ups (T2)*

T1 and T2	X^2	df	RMSEA	CFI	TLI
Configural	71179.185	8570	0.026	0.981	0.98
Metric	71500.569	8698	0.025	0.981	0.981
Scalar	72087.351	8760	0.025	0.981	0.981

Table 13: *Measurement Invariance between 1-year follow-ups (T2) and 2-year follow-ups (T3)*

T2 and T3	X^2	df	RMSEA	CFI	TLI
Configural	71025.576	8570	0.026	0.98	0.979
Metric	71358.81	8698	0.026	0.979	0.979
Scalar	71938.541	8759	0.026	0.979	0.979

Table 14: *Measurement Invariance between 2-year follow-ups (T3) and 3-year follow-ups (T4)*

T3 and T4	X^2	df	RMSEA	CFI	TLI
Configural	71828.904	8570	0.027	0.978	0.977
Metric	72177.352	8698	0.027	0.978	0.978
Scalar	72768.437	8759	0.027	0.978	0.978

Table 15: *Measurement Invariance between baseline (T1) measures and 2-year follow-ups (T3)*

T1 and T3	X^2	df	RMSEA	CFI	TLI
Configural	68684.742	8570	0.025	0.98	0.979
Metric	69166.876	8698	0.025	0.98	0.979
Scalar	70470.122	8759	0.025	0.979	0.979

Table 16: *Measurement Invariance between baseline (T1) measures and 3-year follow-ups (T4)*

T1 and T4	X^2	df	RMSEA	CFI	TLI
Configural	63658.907	8570	0.025	0.979	0.978
Metric	64230.556	8698	0.025	0.979	0.978
Scalar	66958.163	8759	0.026	0.978	0.977

Table 17: *Measurement Invariance between 1-year follow-ups (T2) and 3-year follow-ups (T4)*

<i>T2 and T4</i>	X^2	df	RMSEA	CFI	TLI
Configural	65102.59	8570	0.026	0.979	0.978
Metric	65612.044	8698	0.026	0.979	0.978
Scalar	67111.424	8759	0.026	0.978	0.978

Appendix C

Findings from measurement invariance for sex can be seen below. Due to only minimal differences in RMSEA values across each model, measurement invariance for sex was confirmed. T1 represents baseline responses, whereas T2 represents the first annual follow-up, T3 as the second-year annual follow-up, and T4 as the third-year annual follow-up.

Table 18: *Measurement invariance for sex at baseline (T1)*

T1	X^2	df	RMSEA	CFI	TLI	SRMR
Thresholds	21240.591	4154	0.026	0.942	0.938	0.058
Loadings	20118.653	4282	0.025	0.946	0.944	0.058
Intercepts	19751.248	4349	0.024	0.948	0.947	0.059
Residuals	17572.328	4416	0.022	0.955	0.955	0.06

Table 19: *Measurement invariance for sex at the first follow-up (T2)*

T2	X^2	df	RMSEA	CFI	TLI	SRMR
Thresholds	20153.062	4154	0.026	0.941	0.938	0.061
Loadings	19015.524	4282	0.025	0.946	0.944	0.061
Intercepts	18313.859	4349	0.024	0.949	0.948	0.062
Residuals	16241.222	4416	0.022	0.957	0.957	0.063

Table 20: *Measurement invariance for sex at the second follow-up (T3)*

T3	X^2	df	RMSEA	CFI	TLI	SRMR
Thresholds	20479.378	4154	0.027	0.935	0.931	0.061
Loadings	19431.803	4282	0.025	0.94	0.938	0.062
Intercepts	18570.207	4349	0.025	0.944	0.943	0.062
Residuals	16370.254	4416	0.022	0.953	0.953	0.063

Table 21: *Measurement invariance for sex at the third follow-up (T4)*

T4	X^2	df	RMSEA	CFI	TLI	SRMR
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Thresholds	19085.454	4152	0.027	0.941	0.937	0.064
Loadings	18231.092	4280	0.025	0.945	0.943	0.065
Intercepts	18015.87	4347	0.025	0.946	0.945	0.066
Residuals	16135.758	4414	0.023	0.953	0.953	0.067

Appendix D

See below for the parcelled items. This process resulted in 14 parcelled items that we used in our latent growth model. They were Int_1, Int_2, Int_3, Ext_1, Ext_2, Ext_3, Neuro_1, Neuro_2, Neuro_3, Som_1, Som_2, Som_3, Det_1, and Det_2. See here for the full list of items in the CBCL (https://nda.nih.gov/data_structure.html?short_name=abcd_cbcl01).

Table 22: *Internalising Item Parcels*

Internalising Items	Int_1	Int_2	Int_3
Fears certain animals, situations, or places, other than school	0.54		
Nervous, highstrung, or tense	0.67		
Too fearful or anxious	0.99		
Worries	0.49	0.31	
Fears they might think or do something bad		0.54	
Feels they have to be perfect		0.79	
Feels too guilty		0.63	
Self-conscious or easily embarrassed		0.28	
Complains of loneliness			0.78
Fears going to school			0.49
Feels worthless or inferior		0.33	0.57

Table 23: *Externalising Item Parcels*

Externalising Items	Ext_1	Ext_2	Ext_3
Argues a lot	0.53		0.42
Demand a lot of attention	0.47		
Easily Jealous	0.51		
Feels others are out to get them	0.47		
Runs away from home	0.36	0.35	
Screams a lot	0.58		
Stubborn, sullen, or irritable	0.73		
Sudden changes in mood or feelings	0.80		
Sulks a lot	0.84		
Suspicious	0.48	0.34	
Temper tantrums or hot temper	0.68		
Whining	0.64		
Cruel to animals		0.57	

Destroys their own things		0.62	
Destroys things belonging to their family or others		0.57	
Lying or cheating		0.49	0.31
Sets fires		0.65	
Steals at home		0.88	
Steals outside the home		0.92	
Vandalism		0.61	
Bragging, boasting			0.65
Cruelty, bullying, or meanness to others			0.67
Disobedient at home	0.34		0.43
Disobedient at school			0.74
Doesn't get along with other kids			0.53
Doesn't seem to feel guilty after misbehaving			0.44
Breaks rules at home, school or elsewhere			0.58
Gets in many fights			0.64
Hangs around with others who get in trouble			0.69
Not liked by other kids			0.36
Physically attacks people	0.30		0.42
Showing off or clowning			0.61
Swearing or obscene language			0.39
Teases a lot			0.63
Thinks about sex too much			0.33
Threatens people			0.48

Table 24: *Neurodevelopmental Item Parcels*

Neurodevelopmental Items	Neuro_1	Neuro_2	Neuro_3
Acts too young for their age	0.36		
Fails to finish things they start	0.62		
Can't concentrate, can't pay attention for long	0.99		
Can't get their mind off certain thoughts; obsessions	0.34		
Can't sit still, restless, or hyperactive	0.82		
Poor school work	0.59		
Inattentive or easily distracted	0.81		
Gets hurt a lot, accident prone		0.68	
Nervous movements or twitching		0.28	
Poorly coordinated or clumsy		0.74	
Prefers being with younger kids		0.33	
Repeats certain acts over and over; compulsions		0.39	
Talks too much	0.34	0.43	
Strange ideas		0.38	0.32
Confused or seems to be in a fog			0.68
Daydreams or gets lost in their thoughts			0.65
Stares blankly			0.90

Table 25: *Somatoform Item Parcels*

Somatoform Items	Som_1	Som_2	Som_3
Nausea, feels sick	0.92		
Stomachaches	0.68		
Vomiting, throwing up	0.79		
Aches or pains (not stomach or headaches)		0.77	
Rashes or other skin problems		0.27	
Other (physical problems without known physical cause)		0.68	
Feels dizzy or lightheaded			0.81
Headaches			0.44
Problems with eyes (not if corrected by glasses)			0.33

Table 26: *Detachment Item Parcels*

Detachment Items	Det_1	Det_2
Would rather be alone than with others	0.68	
Underactive, slow moving, or lacks energy	0.51	
Withdrawn, doesn't get involved with others	1.00	
Refuses to talk		0.74
Too shy or timid		0.75

Appendix E

Below are tables with covariate regression values with psychopathology domains.

Table 27: *Latent growth regressions of pre-term birth with the intercepts and slopes of each factor*

Pre-term Regression	All Participants		Sex: Females		Sex: Males	
Intercepts	P	std.lv (SE)	P	std.lv (SE)	P	std.lv (SE)
General	< 0.001	0.243 (0.006)	0.001	0.236 (0.009)	< 0.001	0.227 (0.008)
Neuro	0.001	0.185 (0.006)	0.396	0.07 (0.009)	0.008	0.191 (0.009)
Internal	0.847	0.012 (0.006)	0.108	0.144 (0.009)	0.454	-0.056 (0.007)
External	0.798	0.011 (0.004)	NA	NA	0.758	-0.018 (0.006)
Detach	0.954	0.004 (0.01)	0.361	-0.096 (0.015)	0.536	0.063 (0.014)
Somat	0.697	0.024 (0.005)	0.104	0.153 (0.008)	0.851	-0.015 (0.007)
Slopes						
General	0.151	0.122 (0.004)	0.306	0.134 (0.006)	0.207	0.142 (0.005)
Neuro	0.008	NA	0.335	NA	0.061	NA
Internal	0.385	-0.084 (0.004)	0.318	0.143 (0.007)	0.218	-0.156 (0.004)
External	0.642	0.054 (0.003)	0.276	-0.156 (0.003)	0.259	0.196 (0.004)

Detach	0.134	0.159 (0.009)	0.233	0.167 (0.014)	0.215	0.22 (0.011)
Somat	0.109	-0.306 (0.004)	0.78	-0.081 (0.006)	0.143	-0.367 (0.005)

Notes: As per table 4, P values represent the probability that the predictive variable (birth weight) is correlated with psychopathology domains. Because we ran these regressions for twelve different models, we took a conservative approach to assessing p values ($0.05/12 = 0.04$), therefore any values below 0.004 were considered statistically significant. Std.lv represents the standardised estimates, indicating the direction of the relationship and to what effect, while standard error (SE) (in brackets) represents the standard error of these values. Negative values in the std.lv column indicate a negative relationship between birth weight and psychopathology (i.e., lower birth weight predicted higher levels of symptoms). Due to the poor fit of the neurodevelopmental slope, standardised estimates of that regression model resulted in errors. We chose to report the p values for the neurodevelopmental slope regardless, despite no values being significant. The errors are likely due to the data not fitting a neurodevelopmental slope very well. Also note that the neurodevelopmental factor is shortened to 'neuro' here. Also, regression of the externalising intercept in females resulted errors, likely due to the extremely low mean values and lack of variance for females in externalising issues. Also, due to a lack of space for formatting, we abbreviated the names of each sub-domain (Neurodevelopmental = Neuro, Internalising = Internal, Externalising = External, Detachment = Detach, Somatoform = Somat).

Table 28: *Latent growth regressions of combined household income with the intercepts and slopes of each factor*

Income	All Participants		Sex: Females		Sex: Males	
Intercept	P	std.lv (SE)	P	std.lv (SE)	P	std.lv (SE)
General	< 0.001	-0.086 (0.001)	< 0.001	-0.081 (0.002)	< 0.001	-0.093 (0.002)
Neuro	0.014	-0.026 (0.001)	0.853	-0.003 (0.002)	0.008	-0.039 (0.002)
Internal	< 0.001	0.121 (0.001)	< 0.001	0.108 (0.002)	< 0.001	0.118 (0.001)
External	< 0.001	-0.091 (0.001)	NA	NA	< 0.001	-0.082 (0.001)
Detach	< 0.001	-0.096 (0.002)	< 0.001	-0.064 (0.002)	< 0.001	-0.113 (0.003)
Somat	0.004	-0.036 (0.001)	0.136	-0.027 (0.002)	0.019	-0.04 (0.001)
Slopes						
General	< 0.001	0.065 (0.001)	< 0.001	0.031 (0.001)	< 0.001	0.085 (0.001)
Neuro	0.038	NA	0.359	NA	0.087	NA
Internal	< 0.001	0.161 (0.001)	< 0.001	0.17 (0.001)	< 0.001	0.126 (0.001)
External	0.001	-0.08 (0.001)	< 0.001	-0.134 (0.001)	0.379	-0.032 (0.001)
Detach	0.001	-0.064 (0.002)	0.077	-0.045 (0.003)	0.009	-0.095 (0.002)
Somat	0.027	-0.088 (0.001)	0.405	-0.05 (0.001)	0.05	-0.105 (0.001)

Notes: As per table 4, P values represent the probability that the predictive variable (birth weight) is correlated with psychopathology domains. Because we ran these regressions for twelve different models, we took a conservative approach to assessing p values ($0.05/12 = 0.04$), therefore any values below 0.004 were considered statistically significant. Std.lv represents the standardised estimates, indicating the direction of the relationship and to what effect, while standard error (SE) (in brackets) represents the standard error of these values. Negative values in the std.lv column indicate a negative relationship between birth weight and psychopathology (i.e., lower birth weight predicted higher levels of symptoms). Due to the poor fit of the neurodevelopmental slope, standardised estimates of that regression model resulted in errors. We chose to report the p values for the neurodevelopmental slope regardless, despite no values being significant. The errors are likely due to the data not fitting a neurodevelopmental slope very well. Also note that the neurodevelopmental factor is shortened to 'neuro' here. Also, regression of the externalising intercept in females resulted errors, likely due to the

extremely low mean values and lack of variance for females in externalising issues. Also, due to a lack of space for formatting, we abbreviated the names of each sub-domain (Neurodevelopmental = Neuro, Internalising = Internal, Externalising = External, Detachment = Detach, Somatoform = Somat).

Table 29: *Latent growth regressions of caregiver education with the intercepts and slopes of each factor*

Caregiver Education	All Participants		Sex: Females		Sex: Males	
Intercepts	P	std.lv (SE)	P	std.lv (SE)	P	std.lv (SE)
General	0.22	-0.008 (0.001)	0.266	-0.013 (0.001)	0.608	-0.005 (0.001)
Neuro	< 0.001	-0.028 (0.001)	< 0.001	-0.046 (0.001)	0.074	-0.02 (0.001)
Internal	< 0.001	0.041 (0.001)	< 0.001	0.08 (0.001)	0.069	0.021 (0.001)
External	0.109	-0.011 (0.001)	NA	NA	0.131	-0.015 (0.001)
Detach	0.602	0.005 (0.001)	0.298	-0.015 (0.002)	0.098	0.025 (0.002)
Somat	0.024	-0.02 (0.001)	0.308	-0.015 (0.001)	0.034	-0.028 (0.001)
Slopes						
General	0.008	0.033 (0.001)	0.003	0.064 (0.001)	0.259	0.021 (0.001)
Neuro	0.149	NA	0.793	NA	0.064	NA
Internal	0.01	0.039 (0.001)	0.002	0.079 (0.001)	0.457	0.015 (0.001)
External	0.13	0.027 (0.00)	0.588	0.015 (0.001)	0.073	0.053 (0.001)
Detach	0.957	-0.001 (0.001)	0.555	-0.014 (0.002)	0.562	0.017 (0.002)
Somat	0.005	-0.083 (0.001)	0.231	-0.064 (0.001)	0.005	-0.118 (0.001)

Notes: As per table 4, P values represent the probability that the predictive variable (birth weight) is correlated with psychopathology domains. Because we ran these regressions for twelve different models, we took a conservative approach to assessing p values ($0.05/12 = 0.04$), therefore any values below 0.004 were considered statistically significant. Std.lv represents the standardised estimates, indicating the direction of the relationship and to what effect, while standard error (SE) (in brackets) represents the standard error of these values. Negative values in the std.lv column indicate a negative relationship between birth weight and psychopathology (i.e., lower birth weight predicted higher levels of symptoms). Due to the poor fit of the neurodevelopmental slope, standardised estimates of that regression model resulted in errors. We chose to report the p values for the neurodevelopmental slope regardless, despite no values being significant. The errors are likely due to the data not fitting a neurodevelopmental slope very well. Also note that the neurodevelopmental factor is shortened to 'neuro' here. Also, regression of the externalising intercept in females resulted errors, likely due to the extremely low mean values and lack of variance for females in externalising issues. Also, due to a lack of space for formatting, we abbreviated the names of each sub-domain (Neurodevelopmental = Neuro, Internalising = Internal, Externalising = External, Detachment = Detach, Somatoform = Somat).